

Can trust induce vertical integration? An experimental study of buyer–seller exchanges with distinct competencies and specific investments

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Abstract

Research has debated the merits of transaction cost and competence-based explanations of make-versus-buy choices. We advance this discussion by arguing that trust not only promotes markets by reducing the hazards of specific investments (as emphasized by previous work), but alternatively triggers vertical integration when parties would like to internalize distinctive competencies and at the same time avoid hierarchical failure. Our experimental results where players choose between markets and hierarchies lend support for this dual role of trust in boundary decisions.

JEL classification: C91, D23, L14

1. Introduction

A central question in strategy research revolves around whether to organize activities in markets (i.e. outsourcing) or hierarchies (i.e. in the firm), with transaction cost economics (TCE) having emerged as a widely articulated theory to grasp such choices. Based on initial ideas by Coase (1937) and Williamson (1985, 1991) explains TCE's fundamental premise as that of firms choosing vertical boundaries based on tradeoffs between organizational costs and market failure; so increasingly hazardous market exchanges induce firms to vertically integrate, while low-risk ones induce them to outsource instead. These hazards, Klein *et al.* (1978) explain, largely involve transaction-specific assets which, since they are worth much less in alternative deals, incite opportunistic parties to hold up or renege on previously agreed promises. At a base level, the choice to vertically integrate helps avert such difficulties, as firms can allocate gains by mandate; although when firms have reasons to *trust*—that is, to believe in the trade partner's reliability,

benevolence, and integrity (Zaheer *et al.*, 1998)—they will opt to outsource instead, since trust mitigates various sorts of market failure and fosters exchange continuity (see Poppo *et al.*, 2008).

Although not denying this view, in this article, we propose a distinct channel through which trust can instead trigger *more*, not less, vertical integration. This alternative hypothesis emerges when, besides the usual market failure arguments, we consider the *distinct competencies* of transacting parties. Prior studies point that firms develop distinctive competencies by specializing in their own craft along the value chain; so, following the logic of comparative advantages (e.g. Langlois and Robertson, 1995; Jacobides, 2008; Colfer and Baldwin, 2016), exchanges can be organized in markets as firms seek to build not only alternative sources of knowledge but also opportunities for trade gains (e.g. Jacobides and Hitt, 2005). However, parties may choose to combine their distinctive competencies in a vertically integrated firm when internal production gains become available from tighter and closer coordination (Argyres and Zenger, 2012; see also Chandler, 1990; Kogut and Zander, 1992; Teece, 2010; Helfat, 2015). So what holds firms from seeking multiple such prospects for vertical integration?

The answer is that hierarchies are known to bring the downside of higher *internal* bureaucratic costs. Without the high-powered incentives of markets, hierarchies can be plagued with agency costs, as internal agents may shirk and reduce productivity-enhancing efforts that may be best brought about in market transactions (e.g. Gibbons, 2005, Puranam *et al.*, 2006, Zenger *et al.*, 2011). We believe trust can have an agency mitigating role, and facilitate vertical integration. For example, GE Aviation's Vice President was inquired about the acquisition of Morris Technologies in 2012, largely as a function of its valuable complex geometry design and precision manufacturing competencies. He justified GE's choice by pointing to the historically trusting and close relationship between the two firms as a significant factor (Aviation, 2012). In other words, trust may not directly cause but perhaps *facilitate* vertical integration as it can increase the perceived value of pooling distinct, specialized competencies. Though usually seen as a leading contributor of transaction cost-based theories of the firm, Coase (2000, 2006) himself has called for an encompassing view of the determinants of firm boundaries. In this view, the decision to acquire or integrate with an exchange partner is crucially influenced by a history of interactions, leading to a desire to strengthen production and capitalize on their trust-based relation (Coase, 2000, pp. 26–27).

This is an intriguing hypothesis generating a different prediction on the facilitating role of trust in vertical integration decisions, though one that unfortunately has not been fully developed conceptually, nor further investigated empirically. A particular difficulty is that empirically, in the real world, it is hard to tease out competencies that are distinct (and so hold the promise of complementarity) from those that are exchange specific. Our experimental methodology addresses this issue. Using structured interactions in a laboratory setting, we pair up 806 buyers and sellers, and have them choose markets or hierarchies under treatments that combine different levels of specific investments, distinct competencies, and trust. Based on the concept of learning curves, we manipulate distinct competencies by allowing sellers to practice the assembly of a Lego® car, with production costs being a direct function of assembly time. Given that trust knowingly reduces opportunism, we argue it can mitigate internal (agency) hazards, so the likelihood of vertical integration to internalize distinct competencies *increases* if partners are perceived to be trustworthy. Running two complementary experiments varying exchange conditions influencing competencies, specific investments, and trust, we find strong support for the hypothesis that trust facilitates vertical integration in a context of distinct competencies.

Our study aims to address an evolving debate on the role of trust in organizational choices. In short, we advance the scholarly debate that whether firms vertically integrate likely depends on a *combined* assessment of both transaction costs and capability-based determinants (Langlois and Robertson, 1995; Jacobides, 2008; Jacobides and Winter, 2012). Specially, we show that trust not only promotes markets by reducing the hazards of specific investments (as emphasized by previous work), but alternatively triggers vertical integration when parties would like to internalize distinctive competencies and at the same time avoid hierarchical failure. Empirically, our experimental design not only improves the assessment of causality, but also implements a controlled exchange setting, with standardized tasks that allow us to uniformly gauge the effect of distinct competencies and separate them from other sources of exchange heterogeneity as well as myriad confounds.

2. Theory background

TCE's case for vertical integration is well articulated. Exchange hazards emerge in the classic holdup problem, where one party makes transaction-specific asset investments that (because they are worth less in alternative deals; Klein

et al., 1978) are subject to a risk of opportunistic ex post exploitation by the partner (who reneges agreed terms; e.g. Williamson, 1985). Given partner behaviors are uncertain and contracts are naturally incomplete, firms vertically integrate to avoid this risk (Klein *et al.*, 1978). A key boundary condition in this scenario is that if contexts involve trust, contractual hazards naturally decrease (Williamson, 1991), therefore leading firms to choose markets. Hence, trust negatively moderates the effect of specific investments on vertical integration, because it lowers holdup risks (Gulati and Nickerson, 2008) and promotes exchange continuity (Williamson, 2008).

Following foundational work by Penrose (1959) and Nelson and Winter (1982), scholars have also examined vertical boundaries based on the logic of heterogeneous competencies (e.g. Jacobides and Hitt, 2005). In contrast to transaction cost logic, the competencies reasoning justifies the choice to internalize an activity based on the relative gains to access productive knowledge within the firm versus outside. The concept of distinct competencies reflects the notion that a firm's resources are superior to those of others, even if such competencies are tradable with many other partners (i.e. non-exchange specific; Gibbons, 2005). Hence, firms with better production competencies in one supply chain stage tend to perform that activity internally (Jacobides and Hitt, 2005: p. 1209) and scan the market to contract with another where they are deficient. In this view, markets emerge as a way to promote exchange gains among specialized agents holding distinct (and complementary) competencies (Langlois and Robertson, 1995; Jacobides, 2008; Colfer and Baldwin, 2016).

But recent organization studies instead suggest that internalizing (rather than outsourcing) distinct competencies may also offer the potential for tighter know-how integration and coordination. For instance, parties may come to share a common identity and language (e.g. Kogut and Zander, 1992; Cremer *et al.*, 2007); manage multiple, synergistic activities subject to scope economies (Chandler, 1990; Teece, 2010); or build systemic platforms exploiting the complementarities of distinct capabilities in the value chain (Helfat, 2015). In these conditions, suboptimal outcomes may materialize in the absence of tight joint problem solving and mutual adaptation (Jacobides and Winter, 2012). Gibbons (2005: p. 19) in fact explains that this alternate view provides “a coherent elemental theory of the firm without specific investments,” describing make or buy choices as a function of enhanced coordination in hierarchies instead of holdup and other potential sources of market hazards.

The internalization of distinct competencies however is not an axiomatic choice. By vertically integrating a supplier, a buyer is subject to potential agency costs; that is, in a hierarchy, low-powered incentives can attenuate the synergic gains of the internalized competencies (Argyres and Zenger, 2012). Agentic issues arise not only due to the complexity of the integration, but from autonomy incentives as well. For instance, one's technical competencies may involve know-how that is difficult to measure and explain, and so need time-consuming alignment (Puranam *et al.*, 2006). All the while, firms must fine-tune the ideal internal autonomy of operations. Typically, sellers in outsourcing relations have extensive autonomy, but end up frustrated as their discretion shortens when brought in house. Therefore, agency-related difficulties can quickly eclipse internal synergies, especially in situations when the expertise of the two firms is highly distinct and specialized. In other words, as opposed to the usual TCE focus on market failure, internal sources of *hierarchical* failure may undermine the integration of distinct competencies.

It is here that we advance our theoretical argument that *trust* helps attenuate this risk of hierarchical failure. Primarily, trust reflects a situation where one is willing to be vulnerable to the acts of another when there is a non-trivial probability of loss due to deliberate cheating or unrealized expectations (Mesquita, 2007). Thus, a transacting party will form an expectation that the other partner will act in a trustworthy manner. Research has shown that these expectations are largely influenced by social conditions surrounding the exchange. One may trust simply out of identifying herself with the goals of another, even if such impressions are based on impromptu *communications* that promote the alignment of goals and expectations. In this latter case, game theorists argue that pre-play interaction allows parties to exchange information and arrive at more beneficial equilibrium outcomes (e.g. Crawford, 1998; Farrell, 1988; Ellingsen and Johannesson, 2004). Parties that many times are mutual strangers can apply informal heuristics such as reciprocity, however flimsy they may be, and evaluate cues on how close their expected gains and personal goals are, coming to perceive the other as reliable or not. Alternatively, trust may emerge from a previous history of interactions, as it happens in *relationally embedded* contexts where parties have existing social attachments (e.g. Granovetter, 1985; Ruef, 2002), and develop mutual social norms of behavior after being acquaintances for a time. Such notions of trust capture ideals of goodwill, honesty, fairness, openness, caring intentions, as well as predictability (Colquitt *et al.*, 2007: p. 910).

To sum up, we model the seemingly opposite roles of trust as a boundary condition in make-or-buy decisions. In contrast to the logic based on the threat of holdup—that trust *negatively* moderates the vertical integration of specific

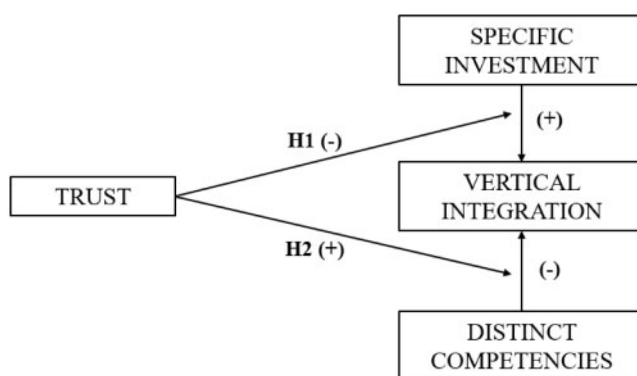


Figure 1. Model summary.

investments—our alternate hypothesis poses that trust will instead *positively* moderate the vertical integration of distinct competencies (Figure 1). To enhance the theoretical precision of our reasoning and provide a more direct connection with our experiments, below we further develop a game-theoretic negotiation model that helps us formally enunciate our hypotheses.

3. Formal model and hypotheses

Let us consider a buyer and a seller (herein “she” and “he,” respectively). We start with the buyer seeking to explore a market opportunity for a given good (in our case, vehicles), so it must decide whether to make or buy it. To get the vehicles produced, she tries to convince a seller to invest in a new plant (which reflects the production technology and capacity needed) by promising to buy the output produced. If the seller makes the investment, buyer and seller have a *market arrangement*. Though it is customary for partners to write a contract at this stage to specify mutual pledges, contracts are incomplete so the buyer may *ex post* renege and demand renegotiation of the price for which the seller would manufacture and deliver the good (Hart and Moore, 1988). Wary of this risk, the seller may refute the initial proposal. The buyer in this case can offer another deal where the seller still builds the plant but immediately sells it to the buyer, while agreeing to assemble the car, for a fixed fee, as an employee in the buyer’s firm. This is equivalent to an acquisition with job hiring, in which the other’s know-how is vertically integrated, so we shall refer to it as a *vertical integration with hiring* (i.e. “with distinct competencies”); that is, the buyer pays for the know-how of the seller and internalizes production as well as the distinct competencies of the seller. This arrangement thus mirrors an internal employment relation, with a previous independent seller hired as an employee of the buyer (e.g. Lafontaine and Slade, 2007). In case both proposals fall apart, we give the buyer a third (fallback) option, in that she can still invest in the plant and produce the good all by herself; a mode which we shall refer to as *vertical integration with own production* (i.e. “without distinct competencies”). In this latter option though, the buyer fails to benefit from the distinct competencies of the seller, and as a single agent has to perform all activities in the production chain. It is obvious that our core theoretical interest lies with the *hiring* option, for it examines the interaction of trust and distinct competencies, but the inclusion of the *own production* fallback option eases trade tensions so ensures the *hiring* choice is not incurred due to a lack of alternatives.

Formally, the buyer can purchase the good produced by the seller at a price p and then resell it for a value of $v > 0$ (net of all selling costs). The production requires an up-front plant investment of $I > 0$, which the buyer wishes the seller will agree to incur. In case the seller refuses the initial proposal of p , the buyer must carry out the investment, which for simplicity we assume it is also I , in all cases: *market* or *vertical integration (with hiring or own production)*. Buyer and seller have respective production costs of c_B and c_S , where, if the seller is specialized in the production of the good, $c_B \geq c_S$. In fact, $(c_B - c_S)$ indicates the extent of *distinct competencies* between buyer and seller. For simplicity, we assume that c_B and c_S are common knowledge, so both parties know their relative efficiencies in producing the good.

3.1 Market arrangement

A *market arrangement* occurs if the seller opts to make the investment for a given pre-specified supply price p (so I is automatically debited from the seller's profit, in our game-theoretic framework). In agreement with Hart and Moore's (1988) idea of incomplete contracts, buyers and sellers can renegotiate p in the subsequent period. Thus, in the next period, buyer and seller profits will be respectively $v-p$ and $p-c_S-I$. We assume $v-c_S-I > 0$, so a positive surplus can be divided between them. If they fail to reach an agreement on p though, the buyer receives zero and the seller pays for the non-redeployable (sunk) part of the investment, $k(0 \leq k \leq 1)$, thereby incurring a loss of $-kI$. In this case, parameter k represents a measure of specific investment; if $k=1$, then the investment is totally non-redeployable. It is instructive to note the difference between investment specificity and distinct competencies in our context: even when sellers are not required to customize their production process to a specific buyer (i.e. $k=0$), they may still be highly skilled (e.g. serving the broader market) and hence be much more efficient than the buyer in the production of the underlying good (i.e. c_S can be much lower than c_B).

In this setting, if specific investments are required, holdup will potentially occur precisely due to the sunk portion of the initial investment. Given a certain price offer from the buyer, in the second period the seller receives $p-c_S-I$ if he agrees with the new price p or $-kI$ if he refuses it. So, the seller viably accepts any renegotiated price $p \geq c_S + I(1-k)$. Opportunistic buyers can thus force a negotiation over the *ex post* surplus $v-c_S-I(1-k)$. Assuming a Nash bargaining solution where they equally split the surplus, buyer and seller finally agree on a price $p = [v + c_S + I(1-k)]/2$. The seller *ex ante* profit then equals $p-c_S-I = [v-c_S-I(1+k)]/2$.

We additionally consider that the degree of opportunism varies according to a parameter $\alpha_B(0 < \alpha_B < 1)$, which measures the risk the buyer will hold up. A highly trustworthy buyer (α_B close to 1) will not attempt to expropriate the seller through *ex post* bargaining, in cases where the seller has committed to a sunk investment. In cases the buyer is highly trustworthy, bargaining will occur over the surplus $v-c_S-I$, which includes the whole up-front investment regardless of its level of specificity. In this case, as α_B approaches 1, the (Nash bargaining) price will converge to $p = (v + c_S + I)/2$ and hence the seller *ex ante* profit equals $p-c_S-I = [v-c_S-I]/2$. Alternatively, in cases where the buyer is highly untrustworthy (α_B close to 0), the outcome will be as described before. Thus, in a *market arrangement*, the equilibrium price, as a function of buyer trustworthiness, equals $p = \alpha_B(v + c_S + I)/2 + (1-\alpha_B)[v + c_S + I(1-k)]/2 = [v + c_S + I-kI(1-\alpha_B)]/2$. Buyer and seller profits, denoted as π_B and π_S respectively, then become:

$$\pi_B = [v-c_S-I + kI(1-\alpha_B)]/2 \text{ and } \pi_S = [v-c_S-I-kI(1-\alpha_B)]/2. \quad (1)$$

3.2 Vertical integration

In contrast to the *market arrangement* situation, *vertical integration* occurs if the seller refrains from investing in the plant, so the buyer herself must invest. In this case, the pair can still negotiate a possible transfer of know-how by the seller whenever $c_B > c_S$. In line with our prior literature review, we assume that vertical integration of distinct competencies facilitates the coordination of interdependent production processes in the value chain. Although for simplicity we do not model these synergies directly, we suppose that in the mode of vertical integration, production costs drop by a factor δ (such that $0 \leq \delta < 1$). In this case, costs become $(1-\delta)c_S$ if the seller does the assembling (i.e., *vertical integration with hiring*) and instead $(1-\delta)c_B$ in the *own production* form of vertical integration. The production synergies from vertically integrating the seller's distinct competencies occur through a hiring agreement where the buyer acquires the seller, hires him to work for a fee x , and incurs the costs of production (i.e., vertical integration *with hiring*). If they fail to reach an agreement on x , the seller earns zero, while the buyer engages in the production process herself, thus receiving $v - (1-\delta)c_B - I$ (i.e., vertical integration with *own production*). We assume that $v - (1-\delta)c_B - I \geq 0$, so it is still profitable for the buyer to invest in the plant and produce the good all by herself, even considering her higher production cost c_B .

If however they do reach a hiring agreement, we expect the following outcome to unfold. Because fee x is fixed, the seller (now an employee of the buyer) will have weaker incentives to work efficiently (even if able to do so). The holdup problem in the market arrangement now turns into an agency problem in the (vertically integrated) firm. However, as before, we assume that the effort exerted by the hired seller will depend on his desire to shirk once employed on a fixed fee basis. We model this effect with parameter $\alpha_S(0 < \alpha_S < 1)$, which captures the extent the seller is perceived to be trustworthy. A highly trustworthy seller (α_S close to 1) will produce at cost $(1-\delta)c_S$, while a highly opportunistic seller (α_S close to 0) will not exert efforts beyond what the *buyer* herself would incur in the

production process, $(1-\delta)c_B$. Thus, the final cost will be $(1-\delta)(\alpha_S c_S + (1-\alpha_S)c_B)$. If the buyer hires the seller for a fee x , she obtains $v - (1-\delta)(\alpha_S c_S + (1-\alpha_S)c_B) - I - x$. Otherwise, by producing herself, she gets (as noted before) $v - (1-\delta)c_B - I$. As the hired seller will accept any $x \geq 0$, both will negotiate over the surplus $\alpha_S(1-\delta)(c_B - c_S)$, which is the difference between the profit the buyer would receive by hiring the seller, net of the seller fee, and the profit that she would receive by producing herself. A Nash bargaining solution then yields $x = \alpha_S(1-\delta)(c_B - c_S)/2$. Hence, in this vertical integration mode, buyer and seller profits are respectively:

$$\pi_B = v - I - (1-\delta)c_B + \alpha_S(1-\delta)(c_B - c_S)/2 \quad \text{and} \quad \pi_S = \alpha_S(1-\delta)(c_B - c_S)/2. \quad (2)$$

3.3 Determinants of vertical integration

Although our context involves three alternative organizational choices—markets, vertical integration *with hiring* (the acquired seller is now an employee of the buyer's assembly line, so assembly involves distinct competencies), and vertical integration *with own production* (i.e. the buyer produces the car herself, so distinct competencies are absent), in this simple formal set-up it is the seller who first decides whether to make the specific investment. The seller will *not* invest in the plant if the market arrangement turns out less profitable than his being vertically integrated. Also, in case $c_B > c_S$, it is not optimal for the buyer to produce herself, as she can either outsource the assembly to the seller or hire him. (In essence, we seek to focus on deals involving the market arrangement or vertical integration with hiring, leaving vertical integration *own production* as a fallback option for the buyer in case negotiations with the seller unravels). Comparing the seller's profits in equations (1) and (2), the seller will not invest in the plant and hence the market arrangement will not happen if:

$$[v - c_S - I - kI(1 - \alpha_B)]/2 < \alpha_S(1-\delta)(c_B - c_S)/2, \quad \text{or} \quad -[v - c_S - (1+k)I] + \alpha_S(1-\delta)(c_B - c_S) - \alpha_B kI > 0. \quad (3)$$

If equation (3) holds, then vertical integration *with hiring* is expected, being that the higher the left-hand side of inequality (3), the higher the likelihood of this vertical integration mode. We can thus immediately check that vertical integration will be discouraged if the term $[v - c_S - (1+k)I]$ is large, indicating that there is a high potential value to be captured in the market arrangement. Notice that this term increases (thus making vertical integration less attractive) if the seller's cost (c_S) decreases, which is aligned with the common view that the presence of more efficient or capable partners encourages outsourcing (Jacobides and Hitt, 2005). Our key terms of interest however are not in the direct effects, but in how trust moderates the make-or-buy choice, or $\alpha_S(1-\delta)(c_B - c_S)$ and $-\alpha_B kI$. The latter term formalizes the known prediction that, with specific investments (i.e. high k), an increase in trust—a growing belief that the buyer will not act opportunistically (i.e. higher α_B)—reduces the value of vertical integration and promotes markets (Williamson, 1991). Notice, however, that this effect depends on trust (α_B). In inequality (3), the total effect of specific investments is $kI - \alpha_B kI$. Thus, in a low trust environment (α_B very small), the last term approaches zero and we should expect the familiar prediction that specific investments should promote vertical integration. In contrast, in a high trust environment (with a high α_B), the effect of specific investments on integration should diminish. Also, note that the effect of α_B in reducing the value of vertical integration is more pronounced when k increases (i.e. with relevant specific investments to be made, trust helps overcome contractual hazards).

In turn, the former term $\alpha_S(1-\delta)(c_B - c_S)$ formally encapsulates our alternate hypothesis that, under distinct competencies, or a high $(c_B - c_S)$, higher trust (i.e. an expectation that sellers will not act opportunistically; that is, high α_S) leads to an *increase* in vertical integration with hiring. Again, the effect is highly dependent on trust; the value of distinct capabilities in a context of vertical integration depends on the expectation that the seller will perform after he is hired in the integrated firm. Said differently, trust increases the confidence the buyer can hire the seller, avoiding agentic opportunism when competencies are internalized. Formally, we state these effects as follows:

H1: An increase in trust (i.e. a rise in the seller's perceptions that the buyer is trustworthy) reduces the effect of specific investments on the likelihood of vertical integration (with hiring).

H2: An increase in trust (i.e. a rise in the buyer's perceptions that the seller is trustworthy) enhances the effect of distinct competencies on the likelihood of vertical integration (with hiring).

4. Experimental methodology

To test our predictions, we conduct two controlled laboratory experiments involving vertical integration choices—the first of which follows the interaction structure of the previous section and tests both H1 and H2, while the second

changes the sequence of decisions and specification of forms of trust to more specifically scrutinize the effect predicted in H2, while abstracting from up-front-specific investments and introducing the role of production synergies (via parameter δ).

A crucial contribution of our design is to separate the effect of specific investments from the effect of distinct competencies, and how these conditions are moderated by trust. Given our controlled setting, we are able to uniformly vary and gauge distinct competencies. Namely, we adopt an assembly task by means of a 41-piece Lego[®] toy (a Lego[®] vehicle, model no. 8122 “Desert Viper,” henceforth “Lego[®]” or “vehicle.” [Supplementary Appendix SA](#) presents an illustration of a participant assembling the Lego[®] toy). The assembly requires both know-how and investment in a (fictitious) plant. The know-how to assemble a Lego[®] is subject to learning curve effects, such that the usual assembly time drops in proportion to one’s sequential practices (in our study, 40% after the second time, 60% after the third, 70% after the fourth time, and so on). We accordingly exploit this fact as a source of distinct competencies, that is, in some sessions, sellers but not buyers practice the assembly. We explicitly show all participants a graph with these statistics to make them aware of the learning curve, so players can judge each other’s competence advantages. In each negotiation session, we compute players’ costs and gains at each phase with experimental points, and only at the end convert these into cash payments. We time the Lego[®] assembly done by our experiment subjects, given that production costs are a function of time.

Early in each session, we read instructions and quiz participants to assure understanding, such that responses were individually checked and corrected to guarantee sufficient knowledge of experiment procedures. Finally, we randomly assigned subjects to buyer and seller roles. We paid subjects cash, including a \$5-dollar show-up fee, plus a variable premium based on performance. Average pay was around \$32 dollars. Sessions typically lasted one hour. More details on the comparison between the experiments, experimental protocol in each case, and all survey instruments are included in the [Supplementary Appendices SB–E](#)).

5. Experiment 1

5.1 Experimental setting

The sequence of decisions in this experiment is directly connected with the former theory model. Thus, we simulate a buyer–seller negotiation where the buyer is said to be an automobile company with a market opportunity for a novel vehicle. By selling this vehicle, the buyer will earn gross revenues of 100 experimental points (i.e. $v = 100$ in the model presented earlier). To get the vehicles produced, the buyer wishes to convince a seller to invest in a new plant, so the seller would supply the completely assembled car to the buyer, at a supply price p that is mutually negotiated. Such plant investment costs 40 points (i.e. $I = 40$). If the seller refuses to invest in the plant, the buyer must make the investment herself, for the same 40 points; in this case, the seller agrees to build the plant and the buyer immediately acquires it, so the seller, now as a hired employee, assembles the car in the buyer’s factory. If the seller refuses both offers, then the buyer can still invest in the plant ($I = 40$) and assemble the car herself. In line with our formal negotiation model, the experiment evolves in three sequential phases ([Figure 2](#)).

5.1.1 Phase 1: seller’s decision to invest or not in the plant

In this phase, the seller decides whether to invest in the plant. Should the seller agree to make the specific investment, he will incur $I = 40$ points, as explained above. In this case, buyer and seller will go to phase 2a, whereby they can renegotiate the final supply price p which the buyer is to pay the supplier for the assembly. Should the seller *not* agree to make the specific investment, they go to negotiation phase 2b.

5.1.2 Phase 2a: negotiation of supply price, for the case where the seller invests in the plant

In this phase, buyer and seller renegotiate face-to-face on price p for at most 10 min. If the renegotiation is successful and the two agree on the same or a new supply price, the experiment moves to phase 3a (market arrangement). Otherwise, with negotiation breakdown, the buyer earns 0 (zero) and the seller has a fallback payoff according to the specific investment treatment (explained below, in phase 3).

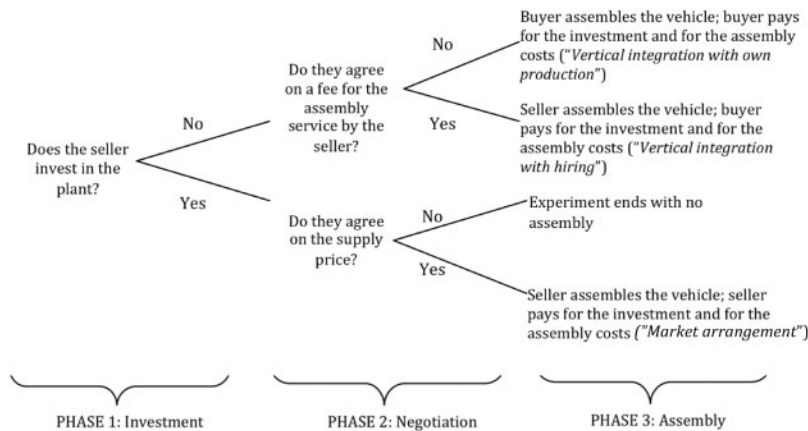


Figure 2. Experiment 1: sequential phases.

5.1.3 Phase 2b: negotiation of the assembly fee, for the case where the seller decides not to invest in the plant

In this case, the buyer herself makes the investment, either by acquiring the seller, or building the plant herself ($I = 40$). In the acquisition mode, the seller, now a hired employee, has to assemble the car for a fixed fee x . Buyer and seller are allotted 10 min to engage in face-to-face negotiations on x . Here however the assembly costs, which are a function of the time the seller takes to assemble the car, will be the responsibility of the buyer. This is done to implement a potential agency problem within the firm: when hired to assemble the car, the seller does not receive the corresponding profits from the operation, thereby potentially reducing his incentives to work efficiently. If they agree on the assembly fee x , then the experiment moves to phase 3b (vertical integration *with hiring*). If however there is no agreement on x , the seller earns 0 (zero), and the experiment moves to phase 3c, where, as a fallback option, the buyer herself can assemble the vehicle (vertical integration *with own production*).

5.1.4 Phase 3: Lego® vehicle assembly

During this phase, either the seller or the buyer will assemble the car, depending on the outcome of negotiation phases 1 and 2. To assess production cost, which is a function of assembly time, we use a stopwatch. Accordingly, assembly cost is equal to an overhead of 5 points, plus 0.10 points for each second taken in assembly (i.e. total cost $c = 5 + 0.10 \times s$). The assembly should take no more than 10 min. In the event the vehicle is not assembled within that time, a session monitor (one of the authors, or an assistant) finishes the assembly and charges a penalty of 40 points. Buyers and sellers assemble the car separately, so buyers do not monitor the seller's assembly activity.

5.1.5 Phase 3 involves three possibilities

In the *market arrangement* (3a), the seller invests in the plant and the seller assembles the vehicle. In this case, the buyer earns the sales price of the vehicle ($v = 100$ points) minus the supply price p paid to the seller. On the other hand, the seller earns the supply price p , minus the assembly cost c (a function of assembly time), minus the investment I (40 points). In the arrangement *vertical integration with hiring* (3b), the buyer acquires the seller, who assembles the Lego®. Here, the buyer earns the price of the car ($v = 100$ pts), minus the assembly fee x paid to the seller, minus the plant acquisition I (40 pts), minus the assembly cost c (a function of assembly time). The seller in turn earns simply the assembly fee x . Finally, in the *vertical integration with own production* mode (3c), the buyer invests in the plant and assembles the car herself. In this arrangement, the buyer earns the price of the vehicle ($v = 100$ pts), minus the cost of assembly c (a function of assembly time), minus the plant investment I (40 points). The seller, on the other hand, earns 0 (zero).

5.2 Treatments

We implement a $2 \times 2 \times 2$ factorial design with three treatments: distinct competencies, specific investments, and exchange conditions affecting perceptions of trustworthiness.

5.2.1 Distinct competencies

Using the Lego[®] assembly task described before, we implement distinct competencies by allowing the seller (but *not* the buyer) to practice assembly at least twice for 15 min. In the treatment involving similar competencies, *both* buyers and sellers have an opportunity to practice assembly. We checked this manipulation by asking buyers to indicate at the end of the experiment the extent they agree with the following statement: “The seller was better trained than me to quickly assemble the car.” The treatment with distinct competencies had a significantly higher score ($P < 0.001$).

5.2.2 Specific investment

In line with the holdup mechanism discussed in the literature (e.g. Klein *et al.*, 1978; Ellingsen and Johannesson, 2004), we manipulate specific investments by the extent to which the seller can recover his previous plant investment if the price negotiation breaks down during phase 2a. In the treatment with high investment specificity, the investment will be totally sunk ($k = 1$) and hence the seller's fallback payoff when there is no price agreement is -40 points. In the treatment without specific investment, the plant is totally salvageable ($k = 0$), so if there is no agreement, the seller loses nothing. We checked this manipulation by asking sellers to indicate their agreement to the statement: “The seller, when investing in the plant, was at risk, given that he could have lost the invested value if he did not receive a good price for the assembled car,” with a higher agreement found in the treatment with specific investment ($P = 0.095$).

5.2.3 Trust

In experiment 1, we examine *exchange conditions* that influence perceptions of the partner's trustworthiness. Our conditions affect distinct stages of the game, and follow established literatures. At a *pre-investment* stage, we manipulate whether buyers and sellers are allowed to meet and talk prior to the actual experimental decisions. Our dummy *personal interaction* equals 0 (zero) if sellers had to decide to invest or not in the plant *without* knowing the buyer's identity *ex ante*, and 1 if sellers could freely interact and communicate with buyers before they decided to invest or not in the plant. In this treatment then, we allowed pre-investment personal interaction to unfold naturally, whereby parties could verbalize their concerns and non-binding promises. Previous theoretical and experimental work argued that pre-play communication promotes an alignment of goals and expectations regarding subsequent negotiation stages (see Farrell, 1988; Crawford, 1998; Ellingsen and Johannesson, 2004), thereby helping reduce (although not totally neutralize) the risk of opportunism. Note that in this treatment, trust is created *before* the seller makes the decision to invest in the plant, so it is expected to affect especially the perceived likelihood that the *buyer* will engage in holdup. In other words, as per H1, the trust in this treatment should have a *negative* interaction with specific investments and so attenuate the effect such investments have on the risk of vertical integration (see Ellingsen and Johannesson, 2004, for a similar empirical study involving holdup and pre-investment personal interaction).

We also *ex post* collected data on a different condition affecting perceptions of trustworthiness: the *relational embeddedness* of buyer–seller pairs (Granovetter, 1985; Ruef, 2002; Lazzarini *et al.*, 2008). Although we do not formally manipulate this condition, we are able to assess the effect of heterogeneous expectations of trustworthiness by adopting Granovetter's (1992: pp. 34–35) notion of relational embeddedness as an exchange condition that influences the parties' perceptions of trustworthiness, insofar as “... how a worker and a supervisor interact is determined not only: by the meaning of these categories in a technical division of labor, but also by the kind of personal relations they have, as determined by a history of interactions.” Accordingly, we measure the relational embeddedness of each randomly assigned pair through a questionnaire administered right after the experimental session. We follow the “social connection” survey proposed by Glaeser *et al.*, (2000: p. 820) in their trust experiment, in which both subjects answer together and try to reach a consensus to avoid measurement error due to relative perceptions of relational strength. (In the second experiment, we will also examine the effect of expectations of each party's trustworthy behavior.) We ask a series of questions to capture the relationship history of the paired subjects as either strangers, acquaintances, or very close friends with recurring interactions. An exploratory factor analysis revealed three items that appear to effectively gauge embeddedness: whether they had met each other before; the extent to which they knew one another well (from “strangers” to “know each other very well”); and the strength of their friendship (from “strangers” to “close friends”). Cronbach's α is 0.869, so we conclude they have a high degree of internal validity. We then averaged the items to create the score metric *embeddedness*.

Different from the previous case, this measure of trust is expected to affect especially the perceived likelihood that the *seller* will shirk in the assembly process *after* the decision to invest or not in the plant. After the initial investment stage, buyers and sellers meet to renegotiate price x , taking into consideration their preferences as well as perceived trustworthiness on the partner, as affected by the level relational embeddedness. Thus, as per H2, trust (*embeddedness*) should *positively* interact with the treatment *distinct competencies*, that is, distinct competencies will more positively affect vertical integration when buyer–seller pairs are in a high embeddedness condition.

5.3 Subjects

Our subjects are graduate (MBA) and undergraduate students from one public and two private universities. The majority is male (63.1%), with an average age of 26.7 years. In experiment 1, we paired up 350 subjects as buyers and sellers, for a total of 175 pairs. We randomly assigned subjects to each treatment cell, and to their roles as buyer or seller. Students were told to negotiate and maximize their own gains, but we gave no instructions as to whether they should deceive or play fairly. Because we have diverse students in our sessions, randomly assigned to distinct treatments, we are able to control for the potential effect on their previous experience, including previous work in organizations. Namely, we add in our regressions variables coding their age and course (MBA or undergraduate). This procedure increases our confidence that our results hold for a broad range of personal characteristics.

6. Results

We test our hypotheses using multinomial logit models, given the three possible organizational arrangements: the *market arrangement* (baseline option), vertical integration *with hiring* (i.e. alternate option, where the buyer acquires then hires the seller), and vertical integration *with own production* (i.e. the fallback option, where the buyer invests in the plant and assembles the car). In our regressions, we include not only the treatments but also a set of controls related to buyer and seller traits potentially affecting outcomes (age, gender, and graduate). These help us to control for remaining demographic differences across experimental conditions. We also estimate robust standard errors clustering by the location where each experimental session was run. Table 1 presents descriptive statistics and frequencies of organizational choices, while Figure 3 shows the percentage of choices according to distinct exchange conditions (the baseline choice is the market arrangement). Table 2, in turn, shows regression results.

Though our central interest lies with moderating effects of specific investments and distinct competencies with trust, as per Figure 1, we start with analyses of main effects, in models 1 and 2 (Table 2). In these models, the coefficient of *specific investment* is not significant, while the estimate for *distinct competencies* indicates that this condition discourages *vertical integration with own production* ($P < 0.05$)—which is expected, given this is a fallback case where buyers would spend significantly more time assembling the car by themselves. Model 2 introduces the trust variables, showing opposite effects, both of which align with our hypotheses. While *personal interaction* largely reduces the likelihood of vertical integration (either with *hiring* or *own production*, $P < 0.01$), the effect of *relational*

Table 1. Experiment 1: descriptive statistics

	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Market	0.674	0.470	1.00												
2. VI hiring	0.246	0.432	−0.82	1.00											
3. VI own	0.080	0.272	−0.42	−0.17	1.00										
4. Distinct competencies	0.491	0.501	0.02	0.08	−0.16	1.00									
5. Specific investment	0.486	0.501	−0.06	0.08	−0.03	0.03	1.00								
6. Personal interaction	0.526	0.501	0.12	−0.12	−0.02	−0.05	−0.06	1.00							
7. Relationally embedded	0.081	0.213	−0.12	0.15	−0.04	0.01	0.16	0.18	1.00						
8. Seller male	0.663	0.474	0.10	−0.04	−0.10	−0.24	−0.03	−0.19	0.10	1.00					
9. Seller graduate	0.054	0.226	−0.12	0.11	0.02	−0.07	−0.07	−0.20	0.04	0.06	1.00				
10. Seller age	21.316	5.026	−0.12	0.08	0.07	−0.10	−0.04	0.01	−0.08	−0.09	0.47	1.00			
11. Buyer male	0.600	0.491	−0.07	0.14	−0.10	−0.06	−0.09	0.00	0.00	−0.04	0.04	−0.11	1.00		
12. Buyer graduate	0.041	0.200	−0.05	0.09	−0.06	0.10	0.03	−0.22	0.07	0.09	0.21	−0.02	−0.07	1.00	
13. Buyer age	20.988	4.967	−0.09	0.15	−0.09	−0.07	−0.02	−0.11	−0.06	0.04	0.05	−0.10	0.03	0.41	1.00

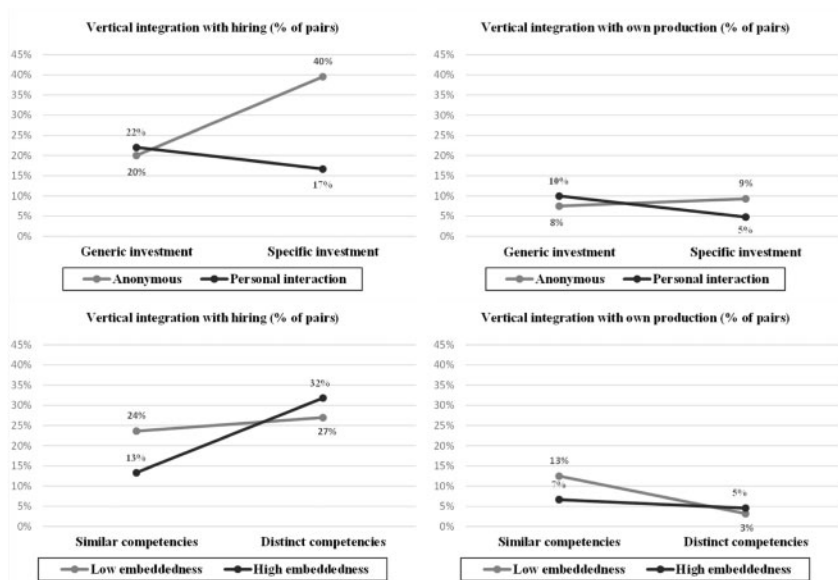


Figure 3. Experiment 1: summary of choices according to exchange conditions.

embeddedness is positive and promotes *vertical integration with hiring* ($P < 0.10$). These distinct effects can be explained by the fact that each of these trust variables mitigates distinct exchange hazards as explained before. The effect of *personal interaction* occurs at a pre-investment stage, so promotes higher investment in the specific plant, as sellers perceive lower holdup hazards in the *market* arrangement. In contrast, the effect of *relational embeddedness* occurs at a post-investment stage, after which parties freely negotiate the terms of trade and buyers can decide to hire their paired sellers or alternatively engage in own production (if they had not previously invested in the plant). Hiring the seller, as we argued, entails a risk of hierarchical failure instead, if the seller shirks in the production process; but trust (embeddedness) lowers this risk, creating an exchange condition that triggers vertical integration with hiring.

Yet formal testing comes by way of their interactions, that is, by examining how our trust variables *moderate* the effect of *specific investment* and *distinct competencies*. For instance, the non-significant effect of vertical integration in our previous regressions could be due to our previous theoretical prediction that its effect is highly contingent on trust, which is heterogeneous across our treatment cells. Following [Hoetker's \(2007: p. 336\)](#) recommendation to examine interactions in non-linear discrete choice models, we implement split-sample regressions, through which we compare the main effects of our key trust variables across subsamples with and without each treatment. Accordingly, we ran models 3a, b, 4a, and b by splitting our sample in treatments involving the absence or presence of specific investments and distinct competencies, respectively, and examine the expected effects of the relevant trust variables operating in each case (i.e. pre-investment *personal interaction* in the case of specific investments and *relational embeddedness* in the case of distinct competencies).

As predicted, models 3a and b show a *negative* interaction between *specific investment* and trust (in the treatment *personal interaction*). Comparing the coefficients of *personal interaction* in subsamples 3a (non-specific investment) and 3b (specific investment) for the option of *vertical integration with hiring* we observe the effect is significantly more negative in the latter ($P < 0.01$ as per a Chi-square comparison of coefficients). This result is consistent with the visual pattern depicted in [Figure 3](#) and also with our previous theory discussion, where we argued that trust should more likely reduce the propensity of vertical integration when investments are highly specific. For the fallback option of *vertical integration with own production*, *personal interaction* shows a negative effect only in the condition

Table 2. Experiment 1: multinomial logit regression results modeling the likelihood of vertical integration with hiring (VI hire) and with own production (VI own)

	Full sample				Split sample—specific investment				Split sample—distinct competencies			
	(1)	(2)			Absent (3a)		Present (3b)		Similar (4a)		Distinct (4b)	
	VI hire	VI own	VI hire	VI own	VI hire	VI own	VI hire	VI own	VI hire	VI own	VI hire	VI own
Hypothesized variables												
Specific investment	0.503 (0.887)	−1.872 (0.767)**	0.410 (0.526)	−1.804 (0.745)**								
Distinct competencies	0.464 (0.827)	−0.533 (0.839)	0.256 (1.047)	−0.649 (0.911)								
Trust												
Personal interaction												
Relational embeddedness												
					−0.333 (0.051)***	0.345 (0.048)***	−1.830 (0.535)**	−1.580 (0.639)**	1.463 (1.763)	−0.275 (0.771)	5.642 (0.419)***	0.956 (2.418)
Controls												
Seller is male	−0.211 (0.133)	−1.502 (0.709)**	−0.608 (0.035)***	−1.433 (0.690)**	−0.814 (0.403)**	−1.057 (0.971)	−0.328 (0.788)	−1.035 (0.798)	−2.394 (0.462)***	−1.859 (0.954)*	0.896 (0.092)***	−0.258 (1.346)
Seller is graduate	0.543 (1.529)	0.544 (0.124)***	−0.470 (1.892)	0.630 (0.139)***	0.094 (2.460)	−14.040 (1.519)***	15.641 (1.145)***	17.149 (1.148)***	3.011 (2.082)	2.353 (1.130)**	−3.684 (0.440)***	1.676 (1.302)
Seller age	0.061 (0.058)	0.028 (0.010)***	0.113 (0.084)	−0.030 (0.054)	0.064 (0.161)	−0.108 (0.055)**	0.118 (0.098)	0.126 (0.112)	0.02 (0.142)	−0.042 (0.090)	0.107 (0.022)***	−0.235 (0.148)
Buyer is male	0.672 (0.333)**	−0.990 (0.168)***	0.733 (0.252)***	−0.991 (0.412)**	0.413 (0.726)	−0.877 (1.557)	1.055 (0.009)***	−0.008 (1.287)	0.142 (0.047)***	−0.407 (0.278)	0.848 (0.337)**	−15.937 (1.154)***
Buyer is graduate	0.785 (0.345)**	−11.755 (1.340)***	0.332 (0.657)	−12.522 (1.802)***	−15.942 (1.411)***	−14.254 (1.011)***	1.597 (1.578)	−14.608 (1.037)***	−15.146 (2.644)***	−14.509 (1.010)***	1.364 (0.161)***	−13.247 (1.950)***
Buyer age	−0.003 (0.108)	−0.182 (0.197)	0.009 (0.104)	−0.134 (0.209)	0.063 (0.050)	−0.209 (0.013)***	−0.054 (0.128)	−0.119 (0.076)	0.005 (0.001)***	−0.066 (0.144)	0.007 (0.065)	−0.278 (0.296)
Constant	−3.167 (2.467)	3.219 (4.064)	−3.695 (2.771)	3.773 (3.012)	−3.516 (3.808)	5.456 (1.145)***	−1.895 (0.036)***	−1.212 (0.807)	−0.688 (3.336)	2.113 (5.824)	−4.642 (1.221)***	8.116 (2.802)***
N	158		155		84		74		80		75	
(pairs)												
Pseudo R ²	0.101		0.147		0.105		0.198		0.189		0.198	
Wald test of model (P)	<0.001		<0.001		<0.001		<0.001		<0.001		<0.001	

Notes: Multinomial logit regressions with robust (Huber–White) random errors in parenthesis, clustered by each location where the experiment was run. The market arrangement is the baseline choice. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

specific plant that could be potentially subject to holdup. Now, the buyer either outsources assembly for a given price or chooses to hire the seller. In line with our previous theoretical discussion, we now also introduce potential production synergies as the distinct competencies are vertically integrated. These synergies create an incentive to pursue vertical integration, but, as before, the gains from integrating distinct competencies depend on perceptions of trust. Thus, the final prediction is identical to our previous discussion leading to H2. (Due to space limitations, we leave the adjusted formal game-theoretical model corresponding to experiment 2 in the [Supplementary Appendix SF](#)) Each experiment session is broken down into three phases, described below ([Figure 4](#)).

7.1.1 Phase 1: the buyer's offer

In this phase, the buyer chooses between two options to get the seller to assemble the car: *market arrangement* or *vertical integration with hiring*. These two modes differ in important ways. In a *market arrangement*, the buyer offers a supply price p , paid to the seller in exchange for the assembled car. If the seller accepts this offer, he invests in the plant and pays for the assembly costs. Alternatively, in the *vertical integration with hiring* arrangement, the buyer pays for the investment, and offers a fee x to hire the seller to work as a regular employee in her firm. As in experiment 1, to ensure the buyer is not totally dependent on the seller, we also offer her a third (fallback) option, *vertical integration with own production*, whereby she can manufacture the vehicle herself.

7.1.2 Phase 2: the seller's choice

In this phase, the seller decides whether to accept the buyer's offer made in Phase 1. If he does, they move on to either phase 3a (*market*) or 3b (*vertical integration with hiring*). If he rejects the offer, they instead move to phase 3c (*vertical integration with own production*). To ensure the seller is not solely dependent on the buyer's bids, we offer him the option to earn five points (plus his show-up fee) to exit the game at this phase. This option reflects an average profit sellers usually earned in our pre-experiment trials, and represents his opportunity costs.

7.1.3 Phase 3: assembly of the Lego®

In this phase, assembly takes place, and the experiment ends. To compute payoffs and performance, we consider production costs as a function of variable (i.e. assembly time) and fixed costs (i.e. overhead). Variable costs involve 0.10 points per second, while fixed costs are 10 and 5 points, respectively in the market and hierarchy modes. We implemented this small difference to manipulate the synergies discussed in our formal model (as captured by parameter δ), and which usually reflects complementarities in complex, interdependent production processes (e.g. [Kogut and Zander, 1992](#); [Cremer et al., 2007](#); [Zenger et al., 2011](#)). Thus, total cost is $c = 5 + 0.10 \times s$ under the *vertical integration* arrangement and $c = 10 + 0.10 \times s$ under the *market* arrangement. Although internal production synergies should incentivize the choice for vertical integration (as we theoretically discussed earlier), our hypothesis center on how trust *further* promotes integration *after* these synergies are *controlled for*; if internal agency hazards are perceived to be high, parties may refrain from choosing vertical integration even when they could possibly exploit cost savings in an integrated production setting.

Note that in the arrangement involving *vertical integration with hiring*, the assembly costs are still a function of the time the seller takes to assemble; but in contrast to what happens in *markets*, they are now the buyer's concern. As in experiment 1, this was done to implement a potential agency risk; that is, within the hierarchy the seller does not receive the corresponding profits from the operation, thus having reduced incentives to work efficiently. To further magnify the potential agency risk, we introduced an opportunity for the seller to engage in an unrelated task that he could voluntarily choose to complete. In this case, for five points, he could fill out an unrelated survey on his career goals. With this task, which takes on average 2 min and it is due *before* the car assembly is complete, the seller can essentially earn extra money at the expense of the buyer who in this hierarchy mode pays for the extra time.

7.2 Treatments

This new experiment involves a 2×3 factorial design with two treatments: distinct competencies and, as in experiment 1, and exchange conditions affecting perceptions of trustworthiness.

7.2.1 Distinct competencies

As in experiment 1, we implemented *distinct competencies* in a supply chain by considering the steep learning curve associated to the task, and allowing sellers (but not buyers) the opportunity to practice assembly of a Lego® toy car for 15 min, prior to making decisions.

7.2.2 Trust

As in experiment 1, we manipulated two *exchange conditions* affecting a trustee's perceptions of the partner's trustworthiness. In a first condition, just like in the first experiment, we varied pre-play *personal interaction*, being that in the *no interaction* cell, we kept randomly paired buyers and sellers in different rooms, with no awareness of their partner's identity, whereas in the *interaction* cell, partners were free to pair up and chat about their goals and intentions before the actual decisions. Notice that, different from experiment 1, the prediction in this case is that personal interaction will *increase* vertical integration, since it is expected to enhance buyers' perceptions of the trustworthiness of the paired sellers.² In a second condition, while preserving anonymity, we allowed an additional form of interaction where sellers were asked to read and sign a *code of conduct* emphasizing expected trustworthy behavior. To guarantee understanding, experimental monitors read the code to *all* participants and asked sellers to hand write their names and sign the code in the same room, with all buyers still present. However, as in the baseline condition, buyers and sellers were still paired anonymously. After sellers signed the code, buyers were taken to a separate room, so that actual decisions and assembly tasks could commence. To avoid characterizing this manipulation as a determinant of contractual incentives, the code was *non-binding*; that is, there was no penalty if the seller signed the code and then cheated in their subsequent interactions. In the personal interaction treatment, sellers also signed the code of conduct.

In experiment 2, we also more directly assessed negotiators' *character trustworthiness* (Gabarro, 1978; Colquitt *et al.*, 2007). This variable is measured as the mean of six items on a 7-point Likert scale, and reflects concepts of honesty, transparency of intentions, justice, benevolence, concern, and dependability (Cronbach's α is 0.880 so we are confident to have high internal validity). We built this measure by adapting scales from prior research in related settings (e.g. Cummings and Bromiley, 1996; McEvily and Tortoriello, 2011). The inclusion of a direct assessment of *character trustworthiness* (in addition to manipulating exchange conditions that induce trust to emerge between partners) makes it for a thorough assessment of how trust affects choices to vertically integrate. Because in this experiment sellers can act opportunistically when hired in the vertical integration mode, we surveyed the *buyers'* perceptions of the seller's character. Also, because another competing explanation is that such synergies may not be unrealized merely due to partners lacking technical skills for the job (Gabarro, 1978; Colquitt *et al.*, 2007), we further add a control variable coding buyers' *confidence in the competence of their partner* (which scholars often refer to as *competence trust*; Gabarro, 1978), measured as the average of three items, each on a 7-point Likert scale, regarding subjects' expectations about the assembler's abilities, dexterity, and aptitudes with Lego® toys. Our metric of competence trust also shows high internal validity (Cronbach's $\alpha = 0.884$).

A battery of manipulation checks strongly confirms the validity of our design. In a post-experimental survey, we asked players about their level of agreement with the following statement: "In our pair, the person playing the role of seller was more capable of speedily assembling the toy than the one playing the buyer." A mean comparison test reveals a significantly higher score in the *distinct* versus the *similar competencies* cells, for both buyers and sellers ($P < 0.01$). We also checked whether sellers who had an opportunity to practice assembly before the experiment in fact performed quicker assemblies than sellers in the experimental condition with similar competencies. A mean comparison test confirms the latter took significantly more time to assemble the cars ($P < 0.01$). Finally, mean comparison tests indicate that buyers' scores for *confidence in the competence of the partner* are significantly higher in the treatment *distinct competencies* ($P < 0.05$), which again indicates that our manipulations were highly effective in creating competence differentials for sellers in that group.

As for the trust treatments, we also asked in this final survey if buyers agreed sellers had "fulfilled the spirit of the code of conduct." Buyers in the *code of conduct* and *personal interaction* treatments showed notably higher

2 In the first experiment, personal interaction occurred at the investment stage and helped increase *sellers'* perceptions of the trustworthiness of the *buyers*—consequently, reducing their concerns that their specific investments would be devalued in a hostage bargaining situation, in a market arrangement.

Table 3. Experiment 2: descriptive statistics

	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Market	0.370	0.484	1.00													
2. VI hiring	0.351	0.479	−0.56	1.00												
3. VI own	0.279	0.450	−0.48	−0.46	1.00											
4. Distinct competencies	0.494	0.502	0.02	0.15	−0.18	1.00										
5. Personal interaction	0.338	0.474	−0.09	0.25	−0.17	0.01	1.00									
6. Code of conduct	0.318	0.467	0.05	−0.12	0.07	−0.01	−0.49	1.00								
7. Goodwill trust	4.665	1.338	−0.22	0.24	−0.02	0.04	0.44	−0.04	1.00							
8. Seller male	0.565	0.497	0.24	−0.15	−0.10	−0.16	−0.07	0.07	−0.11	1.00						
9. Seller graduate	0.058	0.235	−0.02	−0.01	0.03	−0.02	−0.12	0.07	0.02	0.05	1.00					
10. Seller age	21.007	5.241	0.02	−0.10	0.09	−0.04	−0.15	0.06	−0.01	−0.03	0.46	1.00				
11. Buyer male	0.610	0.489	0.03	0.03	−0.07	0.04	0.12	−0.05	0.02	0.19	0.03	−0.08	1.00			
12. Buyer graduate	0.065	0.247	−0.09	−0.03	0.13	−0.05	−0.08	0.05	−0.09	0.02	0.05	0.08	−0.11	1.00		
13. Buyer age	21.278	4.561	0.04	−0.06	0.02	−0.04	−0.04	−0.06	−0.02	0.07	0.14	0.28	0.14	0.29	1.00	
14. Confidence competence	4.686	1.365	−0.18	0.21	−0.03	0.16	0.34	0.00	0.49	0.08	0.03	0.16	−0.02	0.02	1.00	1.00

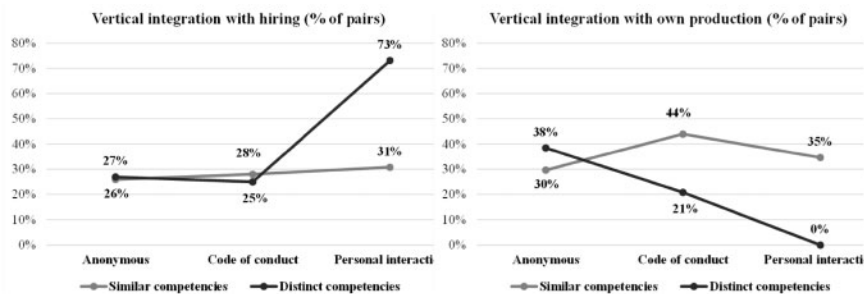


Figure 5. Experiment 2: summary of choices according to exchange conditions.

agreement on this scale than those in the baseline condition where no code was applied ($P < 0.01$). More importantly, buyers perceived code-signatory sellers as displaying higher *character trustworthiness* ($P < 0.01$), while these sellers also less frequently cheated, that is, they more likely refrained from engaging in the unrelated task at the buyer’s expense ($P < 0.05$). We thus conclude that that our trust manipulations (both at the personal and group levels) were also highly effective.

7.3 Subjects

We recruited 456 subjects to experiment 2 (graduate and undergraduate students from the same universities), and randomly assigned them to the treatments as well as the roles of buyer or seller. We used 308 graduate (MBA) and undergraduate students (154 pairs) in main sessions, and other 148 (74 pairs) in extra sessions designed to assess assembly time (to be explained later). Our choice for these subjects follows recent research indicating this population to be a reliable source of data, similar to that of professional managers (Fr  chette, 2015). In total, 58.8% of our subjects are male and, on average, 21 years of age.

8. Results

Table 3 brings descriptive statistics and correlations of key variables, while Figure 5 shows graphically the percentage of organizational choices according to experimental conditions. Part of the relative increase in the incidence of vertical integration, compared with experiment 1, can be explained by the presence of potential production synergies in

Table 4. Experiment 2: results modeling the likelihood of vertical integration with hiring (VI hire) and own production (VI own)

	Full sample			Split sample—distinct competencies			Split sample—distinct competencies		
	(1)	(2)		Similar (3a)	Distinct (3b)		Similar (4a)	Distinct (4b)	
	VI hire	VI own	VI hire	VI hire	VI own	VI hire	VI hire	VI own	VI own
Hypothesized variables									
Distinct competencies	0.055 (0.205)	−0.978 (0.214) ^{***}	0.023 (0.139)		−0.914 (0.232) ^{***}				
Trust									
Personal interaction	0.671 (0.103) ^{***}	−0.701 (0.386) [*]		0.052 (0.165)	0.081 (0.058)	1.416 (0.257) ^{***}		−18.909 (1.053) ^{***}	
Code of conduct	−0.213 (0.383)	−0.322 (0.508)		0.223 (0.376)	0.005 (0.724)	−0.268 (0.407)		−2.384 (0.961) ^{**}	
Character trustworthiness									
				0.381 (0.033) ^{***}	0.187 (0.167)		0.199 (0.050) ^{***}	0.407 (0.114) ^{***}	−0.046 (0.306)
Controls									
Seller is male	0.213 (0.200)	0.224 (0.145)	0.207 (0.272)	0.635 (0.087) ^{***}	0.201 (0.208)	−0.155 (0.277)	0.493 (0.263) [*]	0.284 (0.295)	0.451 (0.179) ^{***}
Seller is graduate	0.942 (0.585)	1.455 (0.217) ^{***}	0.914 (0.384) ^{**}	0.116 (1.852)	1.510 (0.307) ^{***}	1.705 (0.228) ^{***}	0.360 (1.578)	2.561 (0.421) ^{***}	1.284 (0.303) ^{***}
Seller age	−0.027 (0.012) ^{**}	−0.035 (0.054)	−0.025 (0.023)	0.031 (0.024)	−0.037 (0.054)	−0.035 (0.034)	0.031 (0.045)	−0.235 (0.094) ^{**}	0.011 (0.010)
Buyer is male	−0.981 (0.410) ^{**}	−1.067 (0.536) ^{**}	−0.953 (0.422) ^{**}	−0.594 (0.650)	−1.069 (0.599) [*]	−1.390 (0.169) ^{***}	−0.474 (0.795)	−1.203 (0.406) ^{***}	−0.507 (1.092)
Buyer is graduate	0.809 (0.174) ^{***}	0.148 (0.629)	0.583 (0.195) ^{***}	1.050 (0.236) ^{***}	0.264 (0.637)	0.660 (0.676)	1.124 (0.284) ^{***}	−1.215 (0.682) [*]	1.279 (1.997)
Buyer age	−0.044 (0.021) ^{**}	0.007 (0.046)	−0.051 (0.032)	−0.115 (0.037) ^{***}	0.01 (0.045)	−0.008 (0.062)	−0.144 (0.034) ^{***}	0.051 (0.036)	−0.023 (0.122)
Confidence in the competence of the partner									
				0.112 (0.070)	−0.041 (0.095)		0.081 (0.218)	−0.328 (0.052) ^{***}	0.239 (0.152)
Constant	1.623 (0.239) ^{***}	1.529 (0.826) [*]	−0.406 (0.384)	1.478 (0.976)	0.502 (0.959)	1.123 (0.863)	0.838 (1.904)	4.059 (2.268) [*]	−1.478 (2.087)
N	150		150	74		76	74		76
(pairs)									
Pseudo R ²	0.086		0.084	0.088		0.251	0.114		0.133
Wald test of model (P)	<0.001		<0.001	<0.001		<0.001	<0.001		<0.001

Notes: Multinomial logit regressions with robust (Huber–White) random errors in parenthesis, clustered by each location where the experiment was run. The market arrangement is the baseline choice. ***P < 0.01, **P < 0.05, and *P < 0.10.

the integrated mode (as explained earlier, we control for such synergies, and expect trust *further* affects the likelihood of vertical integration especially in the mode with hiring).

Table 4 presents our econometric results. As in experiment 1, we model the determinants of vertical integration using a multinomial logit specification involving those three organizational options, where the market arrangement is again the baseline. Models 1 and 2 are baseline specifications with main treatment effects and trust variables, whereas models 3 and 4 test our hypothesis (i.e. how the effect of distinct competencies varies with trust) using split-sample regressions. We introduce *character trustworthiness* separately from the variables measuring exchange conditions affecting perceptions of trustworthiness (*personal interaction* and *code of conduct*) because, as explained previously, exchange conditions can affect the bases on which a person trusts another (e.g. in the *no personal interaction* treatment, pairs are not allowed to see each other and hence cannot form trustworthiness beliefs on each other). In other words, exchange conditions and character trust are positively correlated with each other.

Models 1 and 2, focusing on main effects only, show that the condition *distinct competencies* discourages vertical integration with own production ($P < 0.01$), thus suggesting that buyers will try to capture the gains from exchanging with trained sellers—either through the market arrangement or through vertical integration with hiring. We also see that *personal interaction* and *character trustworthiness* strongly increase the likelihood a buyer vertically integrates the seller ($P < 0.01$), which indicates that trust attenuates the hierarchical hazards expected in a vertically integrated production process.

As in experiment 1 however, our main interest lies in the interaction of *distinct competencies* and trust, as assessed through our split-sample regressions. Aligned with our theory and our previous findings, H2 is again supported. Comparing the coefficients of *personal interaction* for *vertical integration with hiring* in subsamples with similar and distinct competencies (models 3a and b), and consistent with the visual pattern shown in Figure 5, we find the coefficient is significantly larger in the latter ($P < 0.01$ according to a Chi-square test of coefficient comparison). Also, *personal interaction* discourages *vertical integration with own production* in the condition “highly distinct competencies” ($P < 0.01$). The coefficient of *code of conduct*, in contrast, has no significant effect on the risk of *vertical integration with hiring*. This result is intriguing, in that surveyed buyers notably perceived code-signatory sellers to be more trustworthy than sellers who did not sign the code. Yet, similarly to the effect of *personal interaction*, the *code of conduct* does appear to negatively influence the choice of *vertical integration with own production* in the cell with highly distinct competencies ($P < 0.05$).

Also supporting H2, a comparison of the coefficients of *character trustworthiness* across models 4a and b for the option of *vertical integration with hiring* confirms that the coefficient is significantly larger in the subsample with distinct competencies ($P < 0.01$). Notice that this result holds even when we add the control variable coding buyers’ *confidence in the competence of the partner* (which captures the idea of trust in the skills, not necessarily the goodwill, of the partner; Davis et al., 1997), which positively affects the likelihood of *vertical integration with hiring* in the subsample with distinct competencies ($P < 0.05$). Although its point estimate is larger than in the subsample with similar competencies, a test of coefficient comparison indicates the difference is not significant at conventional levels. In other words, at least in our context, exchange conditions affecting *character trustworthiness* appear to be relatively more relevant to reduce expectations of hierarchical failure than differences in the expected competence of partners. This result is consistent with our previous discussion identifying agency-based hazards as critical determinants of hierarchical failure under vertical integration—hazards that tend to escalate when buyers perceive sellers as relatively more dishonest. *Confidence in the competence of the partner* however negatively affects the choice of *vertical integration with own production* in the treatment with similar competencies, suggesting—as expected—that in this case buyers would like to exchange with sellers that they somehow perceive to be relatively more competent.

9. Discussion: contributions and future advances

Why do firms make or buy? For a long time, strategy scholars have heavily relied upon TCE to address this question, albeit lately competing explanations in the realm of distinct competencies have gained more traction. In agreement with a growing body of research (e.g. Coase, 2000; Jacobides and Hitt, 2005), we see holdup hazards, as well as other forms of market failure, as relevant but insufficient to fully explain make-or-buy choices. Our contribution to this debate builds off from a competence-based perspective: although specialized agents can interact in market exchanges, they may also pool their distinct competencies within a vertically integrated firm as long as they trust that those competencies will be successfully and efficiently deployed. Our argument follows Coase’s (2000) conjecture

that trust can play a role in vertical integration that is not only distinct from that usually prescribed in TCE studies (i.e. trust reduces vertical integration) but in fact opposite (i.e. in some instances, trust may actually prompt the integration of competent partners). We empirically test and confirm the dual and contingent role of trust through a novel and controlled experimental setting, where we more precisely consider specific assets apart from distinct competencies, and showcase the opposite moderating roles of trust in vertical integration decisions.

Beyond the diplomatic suggestion that both transaction and competence-based explanations are concurrently significant to explain make-or-buy choices, our study offers deeper insights into the micro determinants of vertical integration, in contexts involving boundary conditions of distinct competencies and exchange-specific assets. To begin, we reveal the myriad ways in which various forms of trust can affect boundary choices. Considering that the effect of trust in promoting market exchange has been relatively well studied (e.g. Gulati and Nickerson, 2008, Poppo *et al.*, 2008), our findings suggest that scholars should devote more research effort to examine the effect of trust in promoting the internalization of distinct competencies. In other words, trust may not only act as a mechanism to foster co-operative exchange among more or less autonomous agents (e.g. Granovetter, 1985), but also promote, in another direction, the expansion of hierarchical firms.

What is more, we notice that whether exchanges involve specific investments or distinct competencies determines the distribution of bargaining power in the negotiation, and this brings critical implications for who needs to trust whom. In exchanges where sellers make specific investments they run the risk of becoming hostage in ex post dealings, whereas in those where buyers have a distinct competence disadvantage they may be at the mercy of sellers' shirking tendencies. Our model explicitly distinguishes these matters. In exchanges involving specific investments, it is the seller's perceptions of the buyer higher trustworthiness that triggers a higher tendency for both parties to choose "markets" (i.e. to remain separate). This happens because the seller now feels confident he is less likely to be exploited in ex-post small numbers bargaining in a market arrangement as per Williamson's (1991) predictions. On the other hand, in exchanges involving distinct competencies, it is instead the buyer's perceptions of the seller's higher trustworthiness that triggers a higher tendency for both parties to choose "hierarchy" (i.e. to join forces, in an acquisition with hiring). This happens because the buyer now feels confident she is less likely to be exploited by the seller's agency regarding the engagement with the work to be done, that is, whether the seller will shirk or not. This distinction regarding the directionality of trust then highlights an additional element for the debate of transaction costs versus competence-based explanations for vertical integration (Argyres and Zenger, 2012).

Still on the matter of how trust affects decisions to vertically integrate, we demonstrate that its effect is not simply unidirectional; that is, it is not an axiomatic conclusion that trust induces markets, as early proposed by transaction cost economists. In fact, trust can induce vertical integration too. This is clear in the sign reversal in experiments 1 and 2 for whether participants were allowed to communicate prior to the negotiation stage. The caveat lies in what types of exchanges are at play, that is, who is vulnerable to whom with regards to opportunism (see paragraph above) as well as what types of resources are at stake. In cases where specific investments are at stake, we find that trust induces markets, whereas when distinct resources are at stake, we find that trust increases the likelihood of hierarchies.

In this context, several opportunities exist to expand our findings and methods. We focused on a stylized, dyadic relation that mirrors cases where a party would like to hire or acquire the production capacity of another. In our model, hierarchical failure is based on an agency problem, whereby the hired party has weak incentives to perform once inside the firm, at least compared with an alternative market-based arrangement where profits are more closely linked with effort. Yet, other channels may emerge if we consider alternative sources of hierarchical failure and more generalized exchange settings. It is also the case that to make the directionality of trust even more explicit, further experiments may separate the perceptions of trustworthiness of one party onto the other, in which case new conclusions on "deceit" would become available and have an opportunity to be brought to light more explicitly.

Suppose, for instance, that the acquiring firm is considering a target in the same business segment of an existing division operating for a long time as an internal partner. As discussed by Uzzi (1996), a *paradox of embeddedness* may emerge, in that the internal division, with long-term ties inside the corporation, may be against the acquisition and argue that it is a more trustworthy and competent partner. At first glance, this may suggest that trust will undermine the internalization of external competencies from the potential acquisition target. Recall, however, that we are focusing on whether the *external* (market) partner is perceived as being either more or perhaps less trustworthy. If the acquiring firm trusts the competence and character of the potential target, then the firm will better assess and contrast the potential contributions of the external player *vis-à-vis* the internal division. In this context, an alternative

design could involve multiple potential sellers and buyers, with distinct histories of interaction and competencies, and with an ability to form their own networks (e.g. Lazzarini *et al.*, 2008). Participants can then choose their partners contrasting the competence-based advantages brought by novel relationships and the costs of building new cooperative exchanges.

Another feature of our study is that it is focused on the make-or-buy extremes, without considering hybrid forms such as alliances or long-term relationships. Yet, relational norms in recurring exchanges can not only safeguard against holdup hazards in exchanges involving specific investments, but also help improve coordination of complex tasks (Mesquita and Brush, 2008). In this sense, a potential extension of our study is to implement multiple rounds of interaction, varying the incentives to keep cooperative long-term relations versus the gains from short-term defection. Longitudinal data from repeated interactions could also help unveil dual effects of trust, as they increase perceptions of trustworthiness in market exchanges while at the same time facilitating the integration of distinct capabilities.

Supplementary Data

Supplementary materials are available at *Industrial and Corporate Change* online.

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SUPPLEMENTARY APPENDIX 1 – PICTURE OF THE LEGO® TOY



Note: the authors.

SUPPLEMENTARY APPENDIX 2 – COMPARISON BETWEEN EXPERIMENT

Comparison of the design features and the operationalization of the hypothesized variables in
experiments 1 and 2

	Experiment 1	Experiment 2
Decisions	Pre-investment stage and subsequent negotiation of prices and vertical integration modes (see Figure 1).	No pre-investment stage; more direct price offers associated with alternative organizational arrangements, considering potential synergies in the production process (see Figure 2)
Specific investment	Operationalized as whether the plant investment is salvageable. In the specific investment condition, sellers cannot recover the investment in the plant.	Not examined (there is no pre-investment stage).
Distinct competencies	In the condition involving distinct competencies, only one party is allowed to practice assembly before the experimental decisions. Learning curve effects lead to lower potential costs associated with pre-play training.	
Trust (exchange conditions leading to perceptions of trustworthiness)	• <i>Personal interaction</i> (pre-investment): buyers and sellers know each other and communicate prior to the investment decision. • <i>Relational embeddedness</i>: measure of past interactions between the paired sellers and buyers.	• <i>Personal interaction</i>: buyers and sellers know each other and communicate prior to the actual experimental decisions. • <i>Code of conduct</i>: nonbidding agreement to act in a trustworthy way, signed by sellers. In this experiment, we also directly measured <i>character trustworthiness</i>, by means of survey items of buyers' perceptions of honesty, transparency of intentions, justice, benevolence, concern, and dependability of their paired sellers.

SUPPLEMENTARY APPENDIX 3 – EXPERIMENTS GENERAL SETUP

RESEARCH PROTOCOL – EXPERIMENT 1

1. We welcome participants in the room.
2. Participants receive a document with explanations. The document is read aloud. **(E0)**
3. Participants take a quiz to check understanding and the answers are checked. **(E0)**
4. Division into pairs. Buyers and suppliers are randomly chosen.
5. Sellers and buyers complete the initial questionnaire. **(E1.1)**
6. Training and decision:
 - a.) 15 minutes individual training time to assembly the car.
 - b.) Each buyer and seller pair has 10 minutes to negotiate (i.e., whether price negotiation by assembled car or by fee service).
 - c.) Decision is made between (1) market, (2) vertical integration with hiring, or (3) vertical integration own production, depending on how negotiation above turns out.
7. Assembler (whether the seller or buyer, depending on the vertical integration choice) performs car assembly while being timed.
8. The final questionnaire is completed. **(E1.2)**
9. Calculations and payments to participants are performed.

RESEARCH PROTOCOL – EXPERIMENT 2

1. We welcome participants in the room
2. Participants receive a document with explanations. The document is read aloud. **(E0)**
3. Participants take a quiz to check the understanding and the answers are checked. **(E0)**
4. Division into pairs:
 - a.) Role of sellers are assigned based on prior Lego experience (this to ascertain a clearly separation of skills, reinforced with assembly training step below).
 - b.) Sellers sign the Conduct code. **(E2.3)**
 - c.) Half of the sellers leave the room. Half of the buyers choose the remaining sellers. They talk for 5 minutes.
 - d.) Other buyers are in another part of the room waiting while they complete the initial questionnaire.
 - e.) Sellers that were out of the room return. Pairs are defined and codes are placed on the answer sheet.
5. Sellers and buyers complete the initial questionnaire. **(E2.1)**
6. Training and decision:
 - a.) Sellers train car assembly in the same room but away from buyers for 15 minutes.
 - b.) Buyer receives new bid in the case of *market*. Buyer's decision is individual.
7. Assembler (whether the seller or buyer, depending on the vertical integration choice) performs car assembly with while being timed.
 - a.) In assembly, if the seller accepts the task in *vertical integration with hiring* mode, the personal questionnaire **(E2.4)** must be completed.
8. The new vertical integration (i.e., acquisition) proposal is delivered to the seller.
9. Seller accepts or rejects.
10. The final questionnaire is completed. **(E2.2)**
11. Calculations and payments to participants are performed.

Note: we highlighted in box specific areas which relates exclusively to experiment 1 or 2.

E0 - NEGOTIATION EXPERIMENT

Thank you for participating voluntarily in this experiment of negotiation! You will initially receive 10 experimental points for accepting to participate. Depending on your performance, you will receive another variable amount depending on the number of "points" that you are able to get in your negotiation. In the case that you get a negative result, it will be subtracted from the initial value received. We hope you will do your best to increase your result in this experiment. The instructions were designed to describe the procedures as clearly as possible. At the end of the instructions, you will answer a sequence of questions to ensure full understanding of the procedures before the negotiation itself.

THE NEGOTIATION

The owner of an established automobile company called SPEEDCAR, which has an excellent channel of sales, sees the opportunity to sell a particular model of car. He/she already knows how much he will sell the car for: 100 points. To sell the car, SPEEDCAR must first decide whether to: i. Buy the company from a SELLER, already specialized in the assembly process or ii. Purchase the car (assembled) from this SELLER. In both cases, the SELLER must bear the cost of the investment in the car's assembly line, i.e., pay the equivalent of 40 experimental points. However, in the first case, if SPEEDCAR decides to buy the SELLER'S firm, may choose to pay such a price to cover that investment made by the SELLER. Otherwise, if the SELLER and SPEEDCAR cannot agree on the acquisition of the company, the SELLER shall assemble the car at its premises.

We will simulate the assembly of the car using a 41-piece Lego set. At the time of car assembly, monitors of the experimental session will time the total time taken to assemble it (in seconds). This time will subsequently be converted to experimental points, so that the longer it takes to assemble, the higher the cost (in points) associated with the car production.

[Only in instructions for treatment DIFFERENT COMPETENCIES – EXPERIMENT 1]

Prior to negotiation, the SELLER will have 15 minutes to freely and individually train the car assembly. We suggest that in your training the SELLER try to assemble the car at least twice. At each of these times, the SELLER must fully assemble and disassemble the car. Note that the higher the training, the less time spent on assembly as shown below.

[Only in the instructions for treatment DIFFERENT COMPETENCIES – EXPERIMENT 2]

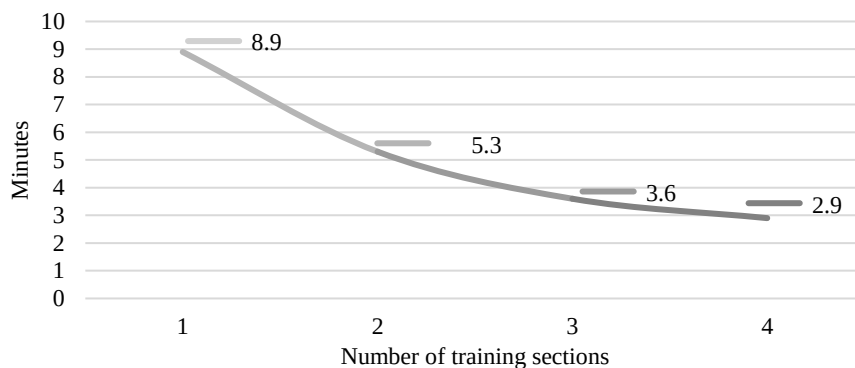
At the beginning of the experiment you will be randomly assigned to the function of SPEEDCAR or SELLER. Once paired, SPEEDCAR and SELLER will remain in separate rooms; one will not know the other's identity throughout the experiment. Prior to their decisions, SELLER and SPEEDCAR will individually complete a questionnaire. However, the SUPRIDOR, prior to the decision, will complete a questionnaire on negotiator characteristics. SPEEDCAR will not know the identity of your SELLER, but will know if it has been paired with a SELLER of a particular "group". In this case, the SELLERS will be separated into two groups: those who have signed a code of good conduct and one that has not signed.

[Only in the instructions for treatment EQUAL COMPETENCIES – BOTH EXPERIMENTS 1 AND 2]

Prior to negotiation, the SELLER and BUYER will have 15 minutes to freely and individually train the car assembly. We suggest that in your training the SELLER try to assemble the car at least twice. At each of these times, the SELLER must fully assemble and disassemble the car. Note that the higher the training, the less time spent on assembly as shown below.

Prior to negotiation, the SELLER will have 15 minutes to freely and individually train the car assembly. SPEEDCAR will not be present at the time of training. We suggest that in your training

the SELLER try to assemble the cart at least twice. At each of these times, the SELLER must fully assemble and disassemble the car. Note that the more training involved, the greater the learning, and therefore the less time (and consequently less cost) spent on assembly, as shown below.



Learning curve in assembly of car: average time vs repetitions

Note: 9 minutes = 540 seconds; 5 minutes = 300 seconds; 3 minutes = 180 seconds.

At the beginning of the experiment you will be randomly assigned the role of SPEEDCAR or SELLER. Then the SELLER and SPEEDCAR will be placed in the same room and SPEEDCAR will be able to choose which SELLER they would like to transact with. Prior to the experiment itself, SELLER and SPEEDCAR will individually complete a questionnaire.

The sequence of decisions will be as follows. Please see the outline at the end of these instructions to view the decisions more globally.

STEP 1: SPEEDCAR's decision to buy the company or buy the car from the seller

At this stage, SPEEDCAR will decide whether to buy the company from the SELLER or purchase the car previously assembled by the seller. Note that in both cases the SELLER will have to invest in an assembly line to manufacture the car. SPEEDCAR, individually, will mark their decision on a form to be distributed to the monitors of the experiment.

If SPEEDCAR chooses to acquire the company from the SELLER, it must offer a price for the company. Similarly, if you choose to purchase the car already assembled, you must offer the SELLER a price for the assembled car.

STEP 2. NEGOTIATION

2.a. SPEEDCAR DECIDES TO BUY SELLER'S COMPANY

SPEEDCAR makes an acquisition bid and the SELLER will decide whether to accept or reject the offer. If the SELLER rejects the proposal, the SELLER will receive 0 (zero) and SPEEDCAR must assemble the cart itself. In this case, SPEEDCAR should invest in the factory, thus paying a value (investment) of 40 experimental points.

If the SELLER accepts the proposed acquisition of the assembly line, the SELLER will receive the price for his company, from which must be subtracted the investment made in the plant (40 points). Then the SELLER will work for SPEEDCAR. In this case, SPEEDCAR will incur assembly costs, depending on the time the SELLER spends to assemble the car.

[Only in the instructions for treatment LOW INVESTMENT SPECIFICITY – EXPERIMENT 2]

Once hired, during the assembly of the car the SELLER will be offered 5 additional experimental points (in addition to what he would normally receive) for him to perform another "service" not related to the production of the car, which he can accept or not. This other "service" will involve completing a questionnaire, which takes about 1 minute. As the cost of assembly will be borne by SPEEDCAR, this possible delay in assembly will reduce the result for SPEEDCAR, but will

not reduce the result of the SELLER; In fact, the SELLER will receive 5 additional points. The total cost, depending on the assembly time, will be paid by SPEEDCAR. The SELLER may or may not complete the questionnaire for additional points.

In this case, SPEEDCAR will receive the resale value of the car (100 experimental points) minus the price paid by the SELLER'S company, less the cost of assembling the car (depending on how long the SELLER takes to assemble the car, as will be explained below); whereas the SELLER will receive the price agreed by his firm minus the initial investment in the factory (40 experimental points) plus the additional 5 points if he agrees to complete the questionnaire. Then the experiment will proceed to Step 3a.

If there is no agreement on the price of the assembled car, SPEEDCAR can still assemble the car by itself. In this case, the SELLER receives 0 (zero) and SPEEDCAR the resale value of the car (100 experimental points) minus the initial investment in the factory (40 experimental points) minus the cost of assembling the car (depending on the time that SPEEDCAR spend on mounting the car as explained below).

[Only in instructions for treatment HIGH INVESTMENT SPECIFICITY– EXPERIMENT 1] If there is no agreement on the price of the assembled car, the SELLER shall still pay the initial investment in the assembly line. Thus, if there is no agreement, the SPEEDCAR will obtain a negative value of -40 experimental points. The BUYER, in turn, will receive 0 (zero).

[Only in the instructions for treatment LOW INVESTMENT SPECIFICITY – EXPERIMENT 1] If there is no agreement, the SELLER will receive 0 (zero). Although the SELLER must still pay the initial investment in the assembly line, in this case it is assumed that he can resell the line in the market at a price equivalent to the initial investment, thus generating a net result of 0 (zero). The BUYER will also receive 0 (zero).

2.b SPEEDCAR DECIDES TO PURCHASE SELLER-ASSEMBLED CAR

If SPEEDCAR makes an offer to purchase the assembled car, the SELLER then decides whether to accept or reject it. If the SELLER rejects the proposal, the SELLER will receive 0 (zero) and SPEEDCAR must assemble the car itself. In this case, SPEEDCAR itself should invest in the factory, thus paying a value (investment) of 40 experimental points.

If the SELLER accepts the proposal, the SELLER will make the investment in the factory (40 points) and this investment can no longer be reversed. Then SPEEDCAR may make a counter-offer to the SELLER, that is, they may change the agreed price. SPEEDCAR may also choose not to make a counter offer, thus maintaining the starting price.

The SELLER may then accept or decline the counterproposal. If they refuse, they will incur the plant cost of 40 experimental points, thus obtaining a negative value of -40 points. If they accept, they will receive the new price offered by SPEEDCAR minus the initial investment in the factory (40 experimental points), less the assembly cost (depending on the time it takes for the SELLER to assemble the car). If SPEEDCAR decides not to change the price, then the SELLER will receive the initially accepted price minus the initial factory investment (40 experimental points) minus the assembly cost. SPEEDCAR will in turn receive the resale price of the car (100) minus the price paid to the SELLER for the car.

STEP 3. Car Assembly

Monitors will time the car assembly process. The assembly cost is defined by the time taken to assemble the cart, plus a fixed cost according to formula $5 + 0.10 \cdot t$, where 't' is the assembly time in seconds. The assembly process should last a maximum of 10 minutes. As shown in the graph above, on average, an inexperienced person takes about 500-550 seconds (approximately 9 minutes) to assemble the car, while a person who has previously trained assembling the car for at

least 15 minutes takes, on average, about 200-250 seconds (approximately 4 minutes). There are three possible outcomes at this stage, depending on the outcomes of previous decisions:

3a. SPEEDCAR acquires the SELLER and the SELLER assembles the car

This will occur if SPEEDCAR decides to acquire the company from the SELLER in Step 1 and the SELLER accepts the offer in Step 2.

Once assembled by the SELLER, the experiment will be terminated. SPEEDCAR will receive: the resale value of the car (100 experimental points) minus the price paid by the SELLER'S company, less the cost of assembling the car. The SELLER will receive the price agreed by his firm, minus the initial investment in the factory (40 experimental points),

[Only in the instructions for treatment LOW INVESTMENT SPECIFICITY – EXPERIMENT 2] plus the additional 5 points if he has agreed to complete the questionnaire in the middle of the assembly.

For example, suppose the SELLER has accepted a price of 42 points for his company. Suppose that the SELLER takes a total time of 270 seconds (...)

[Only in the instructions for treatment LOW INVESTMENT SPECIFICITY – EXPERIMENT 2] if does not accept the proposal to complete the questionnaire in exchange for an additional 5 points.

(...) resulting in the final assembly cost of 32 points ($= 5 + 0.10 * 270$). Then SPEEDCAR will receive $100 - 42 - 32 = 26$ experimental points, while the SELLER will receive $42 - 40 = 2$ experimental points.

[Only in the instructions for treatment LOW INVESTMENT SPECIFICITY – EXPERIMENT 2] Suppose, on the other hand, that the SELLER accepts the proposal to complete the questionnaire in exchange for an additional 5 points, thus causing an increase in assembly time. If this time increases to, for example, 520 seconds, the assembly cost (to be paid by SPEEDCAR) will increase to 57 points ($= 5 + 0.10 * 520$). Thus, SPEEDCAR will receive $100 - 44 - 42 = 14$ points, while the SELLER will receive $42 - 40 + 5 = 7$ experimental points.

3b. SPEEDCAR offers price for SELLER assembled car and the SELLER assembles the car.

This will occur if SPEEDCAR decides to purchase the car in Step 1, and the SELLER decides to invest in the assembly line and accepts the price offered by SPEEDCAR in Step 2.

After the SELLER has made the investment and before starting the car assembly, SPEEDCAR may or may not make a counter-offer. If the SELLER accepts the new offer, the SELLER assembles the car.

Once the assembly is completed, the experiment will be terminated. SPEEDCAR will receive the resale value of the car (100 experimental points) minus the price paid to the SELLER; whereas the SELLER will receive the price he has accepted minus the cost of assembling the car (depending on the time taken) minus the investment value of the assembly line (40 experimental points). This price may be either the price initially offered by SPEEDCAR or - if the SELLER accepts - the price for the counter offer.

For example, suppose SPEEDCAR offered a price of 75 points for the assembled car, without changing the price of the counterproposal, and that the SELLER spends a total of 270 seconds so that the final assembly cost is 32 points ($= 5 + 0.10 * 270$). Then SPEEDCAR will receive $100 - 75 = 25$ experimental points, while the SELLER will receive $75 - 32 - 40 = 3$ experimental points. (Note that by investing in the factory, the SELLER will incur a cost of 40 experimental points in addition to the assembly cost, 32).

Suppose now that SPEEDCAR offers a price of 75 points for the assembled car, which is accepted by the SELLER. In this case, the SELLER will automatically make the 40-point investment, which cannot be reversed. Suppose then that SPEEDCAR decides to make a counterproposal for the price: 59 points, which may or may not be accepted by the SELLER. If the SELLER rejects, he will receive -40 points (negative forty points). If the SELLER accepts, imagine again that the SELLER spends a total of 270 seconds so that the final assembly cost is 32 points ($= 5 + 0.10 * 270$). Then SPEEDCAR will receive $100 - 59 = 41$ experimental points, while the SELLER will receive $59 - 32 - 40 = -13$ experimental points. These -13 experimental points will be deducted from the initial value the SELLER received to participate in the experiment (5 points).

3c. SPEEDCAR assembles the car

This will occur if the SELLER has rejected any offer made by SPEEDCAR. Thus, SPEEDCAR itself should invest in the factory and assemble the car.

Once the assembly is completed by SPEEDCAR, the experiment will be terminated. SPEEDCAR will receive the resale value of the car (100 experimental points), minus the cost of assembling the car (depending on the time taken), minus the investment value of the assembly line (40 experimental points); whereas the SELLER will receive zero (0).

For example, suppose SPEEDCAR spends a total time of 300 seconds meaning that the final assembly cost was 35 points. SPEEDCAR then will receive $100 - 35 - 40 = 25$ experimental points. (Note that because it has invested in the plant, SPEEDCAR will incur a cost of 40 experimental points in addition to the assembly cost). The SELLER will receive 0 (zero) if he initially rejected investing in the factory and would receive -40 points if he decided to invest in the factory but rejected SPEEDCAR's counterproposal.

RESULTS

Once the experimental session is over, the total number of points you have earned will be converted to cash and this value will be added to (or reduced by) your 10 experimental points received at the beginning of the trade. The more points you get; the more cash you will receive. The full cash amount will be paid in full after the session. If you have received a negative amount this should be deducted to the amount initially received.

Before starting the experiment, please answer the questions below, which are intended to check your understanding of the procedures and possible results in the experiment. Monitors of the experiment will check your answers and resolve any questions you may have.

QUESTIONS TO CHECK YOUR UNDERSTANDING

1. Suppose SPEEDCAR decided to purchase the SELLER'S assembled car in Step 1, but the SELLER does not accept SPEEDCAR's offer in Step 2. SPEEDCAR then assembles the car, taking 450 seconds.

- a. How much will the SELLER receive?
- b. How much will SPEEDCAR receive?

2. Suppose SPEEDCAR decides to purchase the SELLER'S assembled car in Step 1 and the SELLER accepts SPEEDCAR's offer of 62 for assembling the car in Step 2. Subsequently, in Stage 3 the SELLER assembles the car in 240 seconds.

- a. How much will the SELLER receive?
- b. How much will SPEEDCAR receive?

3. Assume the same as the previous question, but in this case SPEEDCAR makes a counterproposal of 58.

- a. How much will the SELLER receive?

b. How much will SPEEDCAR receive?

4. Suppose SPEEDCAR decides to acquire the seller company in Step 1 and the SELLER accepts a value of 53 in Step 2 to sell the car assembly company. Subsequently, in Step 3 the SELLER assembles the car in 210 seconds.

a. How much will the SELLER receive?

b. How much will SPEEDCAR receive?

[Only in the instructions for treatment LOW INVESTMENT SPECIFICITY – EXPERIMENT 2] 5. Assume the same as the previous question, but in this case the SELLER agrees to answer the questionnaire and assembles the cart within 250 seconds.

a. How much will the SELLER receive?

b. How much will SPEEDCAR receive?

6. Suppose the SELLER refuses to invest in the assembly line in Step 1 and does not accept the offer by the company. Subsequently, in Step 3 SPEEDCAR assembles the cart in 360 seconds.

a. How much will the SELLER receive?

b. How much will SPEEDCAR receive?

Good luck on the experiment!

SUPPLEMENTARY APPENDIX 4 – EXPERIMENT 1 SPECIFIC DOCUMENTS

E1.1: INITIAL QUESTIONNAIRE

Experimental Session _____ Date _____

Name _____

Please answer all the following questions. Some questions are about facts (e.g. date of birth), and others about your opinions. We hope you answer all the questions, but of course, you are free to skip any questions that you are not comfortable answering. If you wish not to answer a question, please make an indication in the margin indicating that you are skipping the question. Other than this, please do not make any other mark on the questionnaire. Your answers will be considered for statistical analysis purposes only and will not be disclosed.

1. What is your gender? _____ Female _____ Male
2. What ethnicity do you consider yourself to be? _____
3. What is your date of birth? Month _____ Year _____
4. Are you a student of: Under Graduation () Postgraduate () Specify course: _____

What semester of your undergraduate or postgraduate degree are you currently completing?
_____ semester

5. I am always reliable. (Indicate your level of agreement or disagreement with this statement)

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

6. Most people think I'm always reliable. (Indicate your level of agreement or disagreement with this statement)

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

7. Would you say most people... _____ try to be helpful _____ are mostly just looking out for themselves?
8. Do you think most people try to... _____ take advantage of you if they have a chance _____ be fair?
9. Generally speaking, would you say that... _____ people in general can be trusted _____ we need to be very careful when dealing with other people.
10. How often do you lend money to your friends?

More than once a week _____
Around once a week _____
Around once a month _____
On occasion _____
Once a year or less _____

11. How often do you lend personal items to your friends (e.g., CDs, clothes, bikes, etc.)?

More than once a week _____
Around once a week _____
Around once a month _____
On occasion _____
Once a year or less _____

12. How often do you intentionally leave the door of your room or apartment unlocked (when no one is home)?

Very often _____
Often _____
Sometimes _____
Rarely _____
Never _____

Indicate your level of agreement or disagreement with statements 14 to 19.

13. Most people trust each other.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

14. Most people are reliable.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
-----	-----	-----	-----	-----	-----	-----

Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree
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15. Most people are basically honest.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

16. People in general are good and generous.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

17. Nowadays, it's very difficult to depend on someone.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

18. When a person interacts with strangers, it is best to do so with caution until this stranger proves to be reliable.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

E1.2: FINAL QUESTIONNAIRE

Experimental Session _____ Date _____

Buyer Name _____

Seller Name _____

Try to reach an agreement on the answers to these questions. If you absolutely disagree on a particular question, feel free to provide two different answers.

1. Have you met before? ___ Yes ___ No. If yes, continue with parts 1.a and 1.b.

1.a) When did you first meet (month and year) _____?

1.b) Approximately how many times have you talked in the last six months? _____

2. Do you have any extra-curricular activities that you participate in together? ___ Yes ___ No

2.a) If yes, state the number of extra-curricular activities in which both participate: _____

3. Have you two ever been together for social events (except for extra-curricular formal activities)?
___ Yes ___ No

If so, approximately how many times have you been together at social events in the last six months?

4. Are you in (or do plan to be in) the same College of this University? ___ Yes ___ No

5. How would you describe your relationship?

_____ Know each other very well

_____ Know each other well

_____ Know each other

_____ One knows who the other is

_____ Don't know each other

If you both know each other, continue with the questions indicated by letters below; otherwise skip to question 6.

5a.) Have you had any disagreements in the past? (Choose an answer from the list below)

_____ No, never.

_____ Just a little misunderstanding

_____ Occasionally we have minor disagreements.

_____ One or two major disagreements

_____ Many great misunderstandings

5b.) Would you describe your relationship as a friendship? ___ Yes ___ No

5c.) If so, how close is your friendship?

_____ Close friends

_____ Very close friends

_____ Friends

_____ Known

6. How likely are you both to find yourself - intentionally or unintentionally - in one place within the next two weeks? (This includes accidental encounters, such as having a meal in one place).

_____ Very likely

_____ Likely

_____ Unlikely

_____ Very unlikely

7. (TO BE ANSWERED ONLY BY SELLER). "The SELLER, when investing in the industrial plant, was at risk because it could lose its invested value if it did not receive a good price for the assembled car."

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

. (TO BE ANSWERED ONLY BY MANUFACTURER): "The SELLER was better trained than myself to assemble the car quickly"

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

SUPPLEMENTARY APPENDIX 5 – EXPERIMENT 2 SPECIFIC DOCUMENTS

E2.1 - INITIAL QUESTIONNAIRE

Experimental Session _____ Date _____

Name of **Speedcar** Negotiator: _____

Name of **Seller** Negotiator: _____

Please mark your answers using the scale below from 1 to 7 as indicated.

I believe my business partner

i. will be honest during the negotiations.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

ii. will be fair in their negotiations with me.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

iii. will strive to keep their commitments.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

iv. ... will keep their agreements.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

v. ... will not take advantage of opportunities that have come up during the process.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

vi. ... will assemble the car faster than me.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

vii. ... will be better trained than myself to assemble the car.

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Strongly Disagree	Disagree	Slightly Disagree	Neutral	Slightly Agree	Agree	Strongly Agree

E2.2 - FINAL QUESTIONNAIRE

Experimental Session _____ Date _____

Your code in the experiment _____ Name _____

Please answer all the following questions. Feel free to "skip" any questions you may not feel totally comfortable with, but in this case, please make a statement in the margin of the question. Please do not make any other marks on the questionnaire. Your answers will remain completely anonymous, and will only be considered in set for statistical analysis purposes.

1. What is your date of birth? Month _____ Year _____
Are you a student of: Under graduation () Postgraduate () Specify course: _____
2. What semester of your undergraduate or postgraduate degree are you currently completing?
_____ semester
3. What is your gender? _____ Female _____ male
4. What ethnicity do you consider yourself to be? _____

Indicate your level of agreement or disagreement with the statements using the following scale:

Strongly Disagree	1
Disagree	2
Slightly Disagree	3
Neutral	4
Slightly Agree	5
Agree	6
Strongly Agree	7

	1	2	3	4	5	6	7
1. Not honoring commitments after the deal is closed is commonplace and happens.							
2. It's not wrong for someone to try to renegotiate the details of a deal after the deal has been closed.							
3. There is no problem when someone does not honor his promises and commitments, because the "market" offers many other opportunities for alternative transactions, and people should always seek the best for themselves in every transaction.							
4. The cost of building the "reputation" of being a trustworthy person is very high and therefore not worth maintaining.							
5. In general, during negotiations, a person should take advantage of the opportunities that arise, even if it involves cheating.							

During interactions, my partner...

	1	2	3	4	5	6	7
6. ...showed his intentions transparently.							
7. ...fair in his/her conduct and kept the promises made.							
8. ...did not exceed in taking advantage of opportunities that arose							
9. ... was reliable							
10. ...was guided to follow code of conduct							

Throughout the experiment

	1	2	3	4	5	6	7
11. I was able to interact with my partner							
12. In "Outsourced" mode, where the SELLER receives a price bid to deliver the assembled cart, there was a risk that SPEEDCAR would make a lower price bid than the price initially offered.							

In light of what happened in the experiment, please answer ...

	1	2	3	4	5	6	7
13. The person who assemble the Lego was competent and capable							

14. I had confidence in the assembly skills of the person who assemble the Lego							
15. In our pair, the person in the role of SELLER was able to assemble the cart much faster than the person in the role of SPEEDCAR.							

How often do the following events occur in your life? (That is, mark what really happens, not what you would like to happen). Use the scale below:

Never	1
Almost never	2
Rarely	3
Occasionally	4
With some frequency	5
Very often	6
So often I lost count	7

	1	2	3	4	5	6	7
16. Do you lend money to friends or co-workers?							
17. Do you lend personal things to friends or co-workers?							
18. Do you lend valuable things to friends or co-workers?							
19. In your school environment, how often are your valuables (e.g. computer, purse, car keys, etc.) kept in close reach?							
20. "Trust" means putting yourself in situations of vulnerability, where the earnings and well-being of those you trust depend on the goodwill and honesty of the people you trust. Based on this definition, do you "trust" others (whether at work, between friends, or in any environment you frequent)?							

21. Mark an X in the phrase that best matches your experience with Lego toys.
- I. Have never assembled Lego Toys in my Life ()
 - II. I assembled Lego toys a few times during my childhood ()
 - III. I assembled Lego toys many times during my childhood ()
 - IV. Not only did I assemble Lego toys in my childhood... but I still do it today ()
 - V. I am an "ace" in Lego building, and I have a lot of experience ().

E2.3 – CODE OF CONDUCT

COMPANY CODE OF CONDUCT

You are part of a company that values ethical behavior and a sense of collaboration with all those involved in our operations. We try to assemble the best cars as quickly and efficiently as possible. We do not engage in actions that may harm our partners, customers or shareholders (owners). We seek to maintain a reputation of good repute, respect and cooperation in all activities and negotiations with which we engage. All our employees, executives and shareholders follow this guidance.

Signed:

Your name:

(Note: Your name will not be given to the person playing the role of SPEEDCAR; they will only know that you are part of a group of sellers who signed this code of conduct.)

E2.4. – PERSONAL QUESTIONNAIRE

Experimental Session _____ Date _____

Name _____

1. What do you look for in terms of professional development?

2. Why did you choose to take this college course?

3. What are your strengths and weaknesses?

4. Where do you want to be in your career five years from now?

5. What was the most important thing you learned in school?

SUPPLEMENTARY APPENDIX 6 – GAME-THEORETIC FORMALIZATION OF EXPERIMENT TWO

A buyer and a seller (*she* and *he* respectively) have respective positive production costs c_B and c_S , known to both parties. Given the seller is specialized, $c_B \geq c_S$, where $(c_B - c_S)$ indicates the extent of *distinct competencies* between them.

We follow the sequence of decisions in Figure 3, but assume that “nature” starts moving by selecting seller “types,” i.e., whether they are *trustworthy* or not. The initial probability that a buyer finds a trustworthy seller is α , such that $0 < \alpha < 1$; in this, α reflects the extent the buyer’s perceives a seller to be *trustworthy* (which, in turn, can be affected by exchange conditions or bases of trust, i.e., *character-based* trustworthiness). As nature moves and defines buyer and seller types, the buyer must then select between ‘make’ or ‘buy’, as follows.

Organizational Arrangements Chosen by the Buyer

Market arrangement. A buyer offers a seller an outsourcing contract and promises to pay a price p for his car, which the buyer can then resell for a value of $v > 0$ (net of selling costs). For this, the buyer must convince the seller to invest an amount equal to $I > 0$ in a new custom plant dedicated to her conditions, as well as sell the good to her for a price p . Here, the seller also incurs production costs c_S . The seller can reject the offer, but if he accepts it, he automatically invests in the plant and obtains $p - c_S - I$, while the buyer gets $v - p$.

Vertical integration by acquisition. A buyer offers to internalize the seller’s assembly competencies for a price x (i.e. she offers to acquire his firm). If the seller accepts the offer, then he becomes an employee, and the buyer incurs the relevant production costs. For simplicity, assume that the (fixed) wage of the seller is already incorporated in x . Once hired, the seller chooses to exert either high or low effort to assemble the vehicle. If he exerts high effort, the final production cost, paid by the buyer, is c_S (as in the market mode). In case of low effort, the final cost is higher, $c' > c_S$. We consider that if the seller exerts low effort he gets an additional payoff of $\lambda > 0$, which is equivalent to an extra “leisure”. We assume that $\lambda < (c' - c_S)$, i.e., the added final production cost under low seller effort is larger than his (payoff-equivalent)

“leisure”. We also assume that, under vertical integration, the buyer has an added benefit of $\delta > 0$ due to shared expenses and other “synergies” associated with joint production.

Thus, if the seller chooses low effort, he gets $x - I$ (given that, to sell the firm, he must first invest to specialize the plant to the buyer’s demands), whereas the buyer earns $v - c_S - x + \delta$. If the seller exerts low effort however, he attains $x - I + \lambda$, while the buyer gets $v - c' - x + \delta$. There is therefore a risk that, once hired, the seller will incur in a larger production cost than he would otherwise achieve in the market mode where he incurs all relevant production costs. In this setting, we define a trustworthy seller as a seller who always prefers to exert high effort after he sells his firm to the buyer. Untrustworthy sellers, in contrast, always pursue the extra gain λ even if this increases buyers’ costs by $c' - c_S$.

In case the seller rejects any of the offers, the buyer must invest herself in the plant and assemble the vehicle—this is the mode we refer to as *vertical integration with own production*. In this case, given that the buyer’s production cost is c_B , her final payoff will be $v - c_B - I + \delta$, while the seller will get a (normalized) reservation payoff of 0 (zero).

We solve this game using the concept of *Perfect Bayesian Equilibrium*. Here, each player forms beliefs on the types of other players at each decision node based on initial expectations and observed movements. Also, players make the best (payoff-maximizing) decisions based on these beliefs and the strategies of the other players. We thus examine the outcome of buyers’ initial decisions, and derive the equilibria by varying the extent of distinct competencies.

Decisions

Decisions under the market arrangement. The seller always accepts an offer higher than his reservation payoff, which is normalized to 0 (zero). Hence, the buyer will offer $p = I + c_S$. Thus, under the market arrangement, the buyer obtains:

$$\pi_B^M = v - I - c_S. \quad (1)$$

Decisions under vertical integration with hiring. The untrustworthy seller, upon receiving and accepting an offer x , will always choose low effort, given that $\lambda > 0$. So the buyer must expect an untrustworthy seller to always choose low effort, resulting in a production cost

of c' and a seller payoff of $x - I + \lambda$. In contrast, a trustworthy seller will always choose high effort and thus receives $x - I$. Anticipating this, the buyer offers a price for the seller's firm (x) that maximizes her expected payoff. Because all seller types get 0 (zero) if they reject the offer, the buyer must offer at least $I - \lambda$ to get the acceptance of the untrustworthy seller and at least I to get the acceptance of the trustworthy seller.¹ Thus, the buyer payoff if she offers $x = I - \lambda$ is:

$$E(\pi_B^{VI} | x = I - \lambda) = \alpha(v - c_B - I + \delta) + (1 - \alpha)(v - c' - I + \lambda + \delta) \quad (2)$$

In this case, the offer is rejected by the trustworthy seller, which occurs with probability α , thereby leading the buyer to engage in vertical integration with own production, which yields $v - c_B - I + \delta$. In the same vein, the expected payoff if the buyer offers $x = I$ is:

$$E(\pi_B^{VI} | x = I) = \alpha(v - c_S - I + \delta) + (1 - \alpha)(v - c' - I + \delta) \quad (3)$$

Hence, the buyer chooses to offer $x = I$ if $E(\pi_B^{VI} | x = I) \geq E(\pi_B^{VI} | x = I - \lambda)$, or, performing all calculations and rearranging terms:

$$\alpha \geq \lambda / (c_B - c_S + \lambda). \quad (4)$$

If (4) holds, then the buyer offers $x = I$ and her expected payoff is as indicated in (3). For this to happen however, all expected payoffs must be higher than the gain that the buyer would have by offering a very low x (e.g. $x = 0$) and then engaging in vertical integration with own production. Because in this case the buyer has a payoff of $v - c_B - I + \delta$, it is easy to check that a sufficient condition for this payoff to be lower than $E(\pi_B | x = I)$, which is given by (3), is:

$$c_B \geq \alpha c_S + (1 - \alpha)c'. \quad (5)$$

Equilibria

With Distinct Competencies ($c_B > c_S$). For simplicity, suppose that $c_B - c_S$ is sufficiently large so that condition (4) always holds and therefore it is optimal for the buyer to offer $x = I$. If c' is not larger than c_B (i.e. a competent but untrustworthy seller performs better than the less competent buyer), given that $c_B > c_S$ (due to distinct competencies) then (5) will also always hold. Under those conditions, the buyer will choose vertical integration if $E(\pi_B^{VI} | x = I) \geq \pi_B^M$ or, given (1) and (3):

¹ A value between these two extremes is not optimal; for instance, if the buyer offers an x that is slightly below I but above $I - \lambda$, the trustworthy seller will always reject but the untrustworthy seller would accept an even lower price.

$$\alpha \geq 1 - \delta/(c' - c_S). \quad (6)$$

Condition (6) thus confirms that vertical integration by acquisition requires higher perceptions of trustworthiness. The critical level of perceived trustworthiness to make (6) hold will decrease with the extent of synergies under vertical integration (δ) and increase with the added production cost if the task is performed by the untrustworthy seller ($c' - c_S$).

Similar Competencies ($c_B = c_S$). If $c_B = c_S$, then (4) cannot hold, so the buyer offers $x = I - \lambda$. At the same time, the market arrangement can never be optimal: π_B^M , given by (1), is always lower than the payoff obtained under vertical integration with own production, $v - c_B - I + \delta$. Thus, vertical integration by acquisition only occurs if its expected payoff, $E(\pi_B^{VI} | x = I - \lambda)$ is larger than the payoff under own production, i.e., when $E(\pi_B^{VI} | x = I - \lambda) \geq v - c_B - I + \delta$. Given (2), a condition for this to happen is $\lambda \geq (c' - c_B)$. But this is not possible, since with no distinct competencies $c_B = c_S$ and we assumed before that $\lambda < (c' - c_S)$. Therefore, with similar competencies, it is never optimal for the buyer to choose vertical integration by acquisition, and thus this arrangement cannot be an equilibrium of the game.